

Bayesian Dose-Response Modeling for Toxicology Risk Assessment with Application to Antimony Trioxide

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Summary

We estimate the risk function for the dose-response assessment component of the human health risk assessment process using three Bayesian methods: parametric modeling, Bayesian model averaging (BMA), and Gaussian process regression (GPR). The risk function is the relationship that quantifies the probability of an adverse health outcome given varying exposures to an agent (such as a toxin).

We then apply these methods to two datasets: one simulation dataset where we add noise to an underlying known function and one real-world experimental set from a two-year animal study evaluating the impact of antimony trioxide on the development of alveolar/bronchiolar adenomas and carcinomas, offering the first Bayesian dose-response assessment of the agent.

In the Bayesian parametric methods section, we fit nine parametric models motivated by regulatory guidance, past research, and biological plausibility to the data. We also apply semi-informative priors to these models and select the best model using the Bayes Information Criteria (BIC) value.

In the BMA section, we combined our parametric models based on the posterior probability weights to determine the model-averaged risk function. We calculated weights from the posterior probabilities of our models using uniform priors over the model space and bridge sampling, which can efficiently estimate marginal likelihoods.

Finally, we developed a GPR method using the radial basis function (RBF) kernel, incorporating monotonicity and bound constraints to ensure biological plausibility.

We present the strengths, limitations, and areas for further development of these methods. We recommend the BMA approach for estimating the risk function in dose-response assessment, with shape confirmation from GPR. Future work should refine parametric model inclusion, improve prior selection for parameters and over the model space, and develop new approximation methods that reduce computational demand from Bayesian modeling.

1 Introduction

1.1 Dose-Response Assessment

Human health risk assessment is an iterative process that seeks to quantify the hazards associated with exposure to an agent (such as a toxin) on human health across five general steps: problem formulation, hazard identification, dose-response assessment, exposure assessment, and risk characterization (NRC, 2009).

Within this context, dose-response assessment is the process of characterizing the relationship between a particular level or range of exposure to a given agent, previously identified in the hazard assessment, with the incidence or severity of an adverse health effect, such as the development of cancer (NAS/NRC, 1994). For example, researchers may seek to understand the risk of heart attacks given a specific increase in exposure to particulate matter. The methods for characterizing the dose-response relationship differ between cancer and non-cancer outcomes: non-cancer outcomes generally assume a threshold model (i.e., under a certain level, there is no increase in the risk of the event), while cancer outcomes are typically assumed to follow a no-threshold model, where any increase in exposure is associated with an increase in the risk of cancer. This paper concerns cancer dose-response assessment.

Advances in problem formulation, hazard identification, and exposure assessment have allowed for better accounting of personalized environmental toxins and pollutant exposure. These advances are providing richer insights into the depth and variation of exposure. However, the analysis of the dose-response relationship has yet to see similar progress, relying largely on a set of older methods. While such methods have been the foundation of acceptable risk calculations, they fail to adequately incorporate prior knowledge, capture uncertainty around model choice, and are not naturally conducive to probabilistic risk assessment.

Data for characterizing the dose-response relationship is typically derived from epidemiology or toxicology studies, though other methods, such as bioassays, are also used (U.S. EPA, 2012). In experimental toxicology studies, researchers typically expose groups of animal models (such as mice, rats, or Guinea pigs) to a range of doses in a controlled setting. They then evaluate the incidence or measurement of some outcomes, such as an event (e.g., seizure count) or measurements (e.g., liver weight). These experiments are valuable for causal inference. Even though the typical study only tests 2-4 dose groups against an unexposed control with 10-50 animals in each group (Guha et al., 2023), these studies are resource-intensive and, as such, require methods to best characterize the dose-response curve from the available data.

Current procedures for dose-response assessment center on the determination of a point of departure (PoD), the lowest dose at which the response differs from a normal (e.g., control group) response (U.S. EPA, 2012). Depending on the experimental source of the data, additional uncertainty factors may be applied to the PoD to account for additional levels of uncertainty, such as those for limited experiments, extrapolation from animals to humans, natural variation among humans, the mode of action, or if the exposed group is particularly sensitive (such as children) (Chiu and Slob, 2015).

Though there have been other PoD metrics, benchmark dose (BMD) and BMD lower confidence interval (BMDL) are the current preferred methodologies. These are approved and used by EFSA (EFSA Scientific Committee, 2022), OECD countries (2014), the U.S. EPA (2005), and the WHO

(2018). The BMD is typically the dose at which 10% of the test subjects show a response (though other values can be selected) (U.S. EPA, 2012). With a health-safeguarding mandate, regulators typically use the lower confidence interval around the benchmark dose (the BMDL) as the PoD to offer additional caution in evaluating risk. At this point, the dose-response value may be used to extrapolate to lower dose levels based on biologically-plausible models for similar agents. For cancer outcomes, this frequently involves the use of linear low-dose extrapolation.

In the absence of an adequate biological understanding, dose-response assessment is largely a curve-fitting exercise. To address these above challenges and explore new methods to estimate the dose-response function with limited data, we propose a set of Bayesian inference techniques that support probabilistic human health risk assessment from abiotic toxin exposure.

1.2 Risk Function

Current guidelines separate response types into dichotomous/quantal outcomes and continuous outcomes. Our focus will be on dichotomous data, which deals with binary categories (such as "cancer" or "no cancer") to denote the status of the subjects tested in an experiment. These different binary responses are the results of interest in the experiment, and our curve fitting will help draw insights from those. A summary of dichotomous data typically requires three variables: the number of subjects in each dose group, the number of subjects with a given response (e.g., those with cancer), and the dose group (Shao and Shapiro, 2018). The dose is typically evaluated in a standardized form for greater interpretability across different animals and routes of exposure. Researchers often use milligrams per kilogram of body weight per day, noted as mg/kg-bw/day, when discussing doses of toxins. However, other levels (such as micrograms) may also be used.

To model the relationship between exposure to an agent and the development of some adverse event, researchers typically define a risk function f, which characterizes the probability of experiencing the adverse effect $R(d)$ when exposed to a d-level dose. Crump introduced benchmark analysis in 1984, which adjusts for background effects $R(0)$, or the risk at the zero dose. As such, researchers are often interested in the risk above a background response rate, which is excess risk. Here, we will explore the risk function, but we could similarly have developed the excess risk function for dichotomous data using either of the following methods:

$$RE(d) = \frac{R(d) - R(0)}{1 - R(0)} \quad \text{or} \quad RE(d) = R(d) - R(0)$$

Both the risk and excess risk are probabilities. In line with past research, we assume that dichotomous outcomes follow a binomial distribution (Fang, Piegorsch, and Barnes, 2015; Shao and Small, 2011; Shao and Shapiro, 2018) such that,

$$Y_i \overset{\text{indep.}}{\sim} \text{Bin}\left(N_i, R(d)\right)$$

Here, Y_i represents the number of subjects that exhibit the response in category i, N_i represents the number of subjects in group i, and $R(d)$ is the risk at dose d.

1.3 Purpose

There are ongoing calls for regulatory bodies to adopt probabilistic risk assessment to better quantify uncertainty, reduce potentially inaccurate models in risk assessment, and encourage the development of new methodologies for assessing human health risks (U.S. EPA, 2014).

This paper explores three Bayesian methods to estimate the dose-response relationship between an abiotic agent exposure and a response. These methods will supplement the existing literature and policy guidance that focuses on the frequentist approach. Future work will explore how frequentist methods compare to our Bayesian approaches.

Despite the attention in human health dose-response assessment, a single point of departure (or even the lower confidence bound around this value) is insufficient to adequately characterize risk from a given toxin. While current analyses set an acceptable risk level for the population or specific groups, a broader understanding of the relationship between exposure and outcome is useful for understanding disease burden, impact from industrial operations, and the cost-benefit analysis of regulation with varying PoDs. We are interested in this broader relationship, which concerns estimating the dose-response curve for a reasonable range of potential exposure levels.

To do this, we first attempt to fit nine biologically-plausible parametric curves using Bayesian methods. We present the largest number of toxicology-related parametric models fit to a dose-response dataset in the literature. By applying semi-informative priors on the parameters, we combine existing knowledge with observed data to estimate the risk function across the range of tested values and select the best-fitting model based on current practice. This is also called fitting the dose-response curve, and we will use both these terms throughout this paper.

We then provide a novel method for conducting Bayesian model averaging (BMA) for dose-response curves, with a practical example of applying bridge sampling to calculate the marginal likelihood of our models. We then apply uniform priors across the model space and obtain posterior probabilities to weight our parametric model outputs to produce a BMA curve.

Finally, we develop a Gaussian process regression (GPR) with monotonicity and bound constraints to estimate the dose-response function with greater flexibility (than our two previous methods) over the tested dose levels.

We validate all these models by fitting a set of simulated data with added noise to recover the underlying function. We use this simulation to ensure the models can recover a reasonable approximation of the underlying data. Following the simulation, we test a real-world dataset – antimony trioxide – where the underlying relationship is unknown and no official dose-response assessment has been attempted. We present the first set of dose-response curves for this common, high-volume industrial agent.

As noted, PoD values (such as BMD and BMDL) remain important, but understanding the risk function's behavior surrounding the PoD could make these estimates more valuable for additional applications and for the dose-response assessment in its current form. By estimating the dose-response function, we can gain deeper insights into the agent and its effect on individuals and populations. For example, understanding the function over a relevant range of exposures can help classify doses into separate zones, such as high and low doses. These categories can be used to understand areas at which risk is minimal, where it increases the fastest, and the point at which additional exposure may be associated with a leveling off of a response. Additionally, the function may provide new insights into

non-linear effects, countering or supporting the common assumption of a linear no-threshold model for carcinogens.

Understanding the risk function also has practical implications across several domains, such as cost-benefit analysis, individual health, public health, industrial operations, pharmacology and biotechnology development, and policy making. Also, in cumulative risk assessment, the risk function supports estimations for how a confluence of different exposures contributes to individual and population health outcomes. More broadly, an understanding of the curve may support the optimization of experimental design, as key dose ranges may be identified in exploring these curves, which would benefit from further study. At the same time, other areas may be deprioritized, allowing for a more responsible allocation of resources.

We expect to write additional papers on human dose-response assessment to supplement the methods developed here across the human health risk assessment process.

1.4 Antimony

We will apply the methods we develop in this paper to an animal experiment on antimony trioxide, in addition to a simulated dataset.

Antimony is a silver-white crystalline solid metalloid that has been known to humans for at least 10,000 years. Mentioned in 2 Kings 9:30 (NRSV-CE), the chemical has been used for medicinal and cosmetic purposes for at least thousands of years, including by Pliny in antiquity and Jean de Roquetaillade in the 14th century (McCallum, 1999; Patterson, 2021). In the present day, antimony trioxide Sb_2O_3 is the most important antimony compound, as the primary form of the chemical in the atmosphere, a naturally-occurring chemical in many ores, and with uses across multiple high-volume industrial applications (U.S. G.S., 2024).

Antimony trioxide is used as a fire retardant in textiles, plastics, furniture, carpets, and home furnishings (NTP, 2017). It is also an opacifying agent in plastic, glass, and ceramics and a catalyst in polyethylene terephthalate (PET) plastics production. Additional uses include its application in phosphorescent lights and displays and as a component of silicon wafers and diodes for electronics production (NTP, 2017). Antimony alloys (with lead) are used to manufacture batteries, ammunition, and industrial components (Anderson, 2012). Further, antimony is still used in medicine to treat parasites and protozoan infections, such as leishmaniasis (Madusanka et al., 2024).

The main sources of antimony trioxide exposure are industrial settings, such as metal ore smelting, mining, and the production of fire-retardant goods, as well as plastics, glass, and ceramics (NTP, 2017). In the U.S., the occupational exposure limit is 0.5 mg/m^3 (ACGIH, 2015), while the U.S. EPA has an inhalation reference concentration for the chemical of 0.2 μg/m^3 (U.S. EPA, 2014).

Numerous studies have identified potential health risks associated with antimony. In 1989, the International Agency for Research on Cancer identified antimony triode as "possibly carcinogenic to humans," with inadequate evidence from human studies but sufficient evidence in animal studies (IARC, 1989). More recent epidemiological studies have supported a dose-dependent increase in cancer mortality depending on antimony exposure levels, which are summarized in a recent systematic review which found that the compound may be responsible for an increase in lung cancer among workers in high-exposure fields (Saerens et al., 2019).

Beyond epidemiological studies, animal studies have noted a causal linkage between antimony

trioxide exposure and the incidence of various cancers across multiple exposure routes (Schildroth et al., 2022) and have shown a plausible biological mechanism for the chemical's carcinogenic effects. Antimony trioxide depletes glutathione, which disrupts biological processes associated with cellular metabolism and can cause mutations associated with lung tumors and epidermal growth factor receptors (EGFR) (the upregulation of which has been noted in other studies among lung tumors) (NTP, 2017). Furthermore, antimony causes chromosomal damage and an immune response (El Shanawany et al., 2017), further supporting a mechanistically-plausible role in the development of cancer.

Despite these findings, no regulatory body has completed a dose-response assessment for exposure to antimony trioxide. Schildroth et al. (2022) used standard frequentist methods to conduct a dose-response assessment, providing a BMD estimate and an occupational cancer hazard prediction. Their results suggest that current workplace exposure levels are more than 350,000 times higher than the EPA's standard acceptable risk levels (Schildroth et al., 2022).

2 Literature Review

2.1 Bayesian Parametric Approaches

Bayesian methods allow for the incorporation of prior knowledge, greater accounting for uncertainty, and the natural facilitation of probabilistic estimations (Shao and Small, 2011; Chiu and Slob, 2015). The Bayesian approach relies on Bayes' theorem to update the probability of the model parameters based on the observed data and prior beliefs, yielding a posterior distribution. In toxicology studies, data is often sparse, and the inclusion of prior knowledge of model parameters may be beneficial for model fitting (Gelman, Hwang, and Vehtari, 2014).

The U.S. EPA (2022) has identified a set of plausible functions for representing the dose-response curves of toxins, ranging from the relatively simple Logistic and Probit models to more complex models, such as the Weibull and Dichotomous Hill models. Each of these involves a set of assumptions about the form of the dose-response relationship. For example, the Logistic and Probit models assume a linear relationship between the dose and the log odds or Probit of the response and are particularly suitable for binary response data. Conversely, the Weibull model is more flexible and can accommodate the skewed data sometimes found in complex response curves. The Dichotomous Hill and Quantal-Linear models often address data with a threshold effect or are applied to linear low-dose extrapolations. Furthermore, these EPA-recommended models are monotonic, ensuring that the resulting dose-response curve is non-decreasing, a standard assumption in toxicology.

Early work in Bayesian parametric modeling attempted to integrate prior knowledge to improve model reliability in the face of limited or low-quality data (Evans et al., 2001; Axelrad et al., 2005; Woodruff et al., 2007). In recent years, Shao and Small (2011) have explored Bayesian approaches to fitting the Logistic and Quantal-Linear models with non-informative priors on the parameters in their models, and in 2012, expanded on this work to show that dose level selection was an important component of experimental design to reduce uncertainty around BMDL estimates. Davis, Gift, and Zhao (2011) also explored Bayesian approaches to fitting Logistic and Probit models using non-informative priors on the model parameters. In 2012, Shao developed a Probit model that introduced new informative priors at the zero dose level to incorporate some knowledge about the shape of the dose-response curve.

By 2015, Fang, Piegorsch, and Barnes had built on the work of Shao and Small (2011) by reparametrizing the Quantal-Linear model in terms of BMD rather than the standard parameters; this allowed for the incorporation of more intuitive priors that they believed may be shared across similar toxin groups. Fang et al. (2015) expanded this reparameterization to a range of eight parametric models. Despite the apparent desirability of this reparametrized approach, more recent research has not continued with these models parameterized in terms of BMD, potentially because of the additional complexity in the models when reparametrized, leading to additional computational demands or challenges with Markov-Chain Monte Carlo (MCMC) convergence in this form, noted by Fang et al. (2015).

In another significant project, Shao and Shapiro (2018) developed a system for Bayesian BMD estimation that fits multiple parametric models to a given dataset. This Bayesian Benchmark Dose Software (BBMD) supports both dichotomous and continuous data, with eight parametric models for each data type (Shao and Shapiro, 2018). The priors in their analysis were again uninformative, and

model fit was evaluated based on posterior predictive p-values.

Multiple parametric models that could plausibly fit the data present a challenge of model selection. When conducting a dose-response assessment using parametric curve fitting, model selection should be motivated by the underlying biological process (Crump et al., 2010). However, as such knowledge and its relationship to models is often limited, researchers typically fit a selection of models and then choose the best-fitting model by examining fit metrics such as p-value or BIC (WHO, 2018; U.S. EPA, 2022). This best-fitting model is then used as the basis of the risk function and the source of the BMD or BMDL estimates.

A recurring challenge among these studies that fit Bayesian parametric models has been the selection of priors. Researchers have noted that prior choice is particularly important, as certain sets of priors may result in poor MCMC convergence and biased posteriors (Shao and Small, 2011; Fang et al., 2015). In response to these challenges, the U.S. EPA offered a set of priors for continuous and dichotomous data in 2022.

Further challenges with these models concern computational demand, mainly when the dataset is large. Bayesian model fitting typically requires longer runtimes than frequentist modeling (Shao and Small, 2011).

Finally, risk assessments that focus on a single "best" model have demonstrated poor statistical properties, particularly overfitting the data and failing to recover the underlying curve or to produce an accurate BMD estimate (Wheeler and Bailer, 2009; West et al., 2012; Simmons et al., 2015).

More recent advances in Bayesian parametric curve fitting involve extensions such as the combination of parametric models with physiologically-based pharmacokinetic (PBPK) modeling (Chou and Lin, 2020), which seeks to combine the observations and known biological processes in a step beyond the incorporation of prior knowledge about model parameters, but rather about the process beyond the model. Furthermore, Hamza et al. (2021) incorporated a hierarchical framework into their Bayesian parametric modeling to allow for the sharing of information across studies, which allows studies with limited observations but important dose levels (such as the health outcomes of high exposure groups from occupational toxicology studies) to borrow information from animal data with observations across a broader range of exposures.

Despite the challenges, Bayesian parametric modeling has demonstrated superiority in addressing various data structures, accommodating complex dose-response relationships, and providing a probabilistic depiction of risk compared to the frequentist approach (Piegorsch and Bailer, 2005). Given the above challenges, a well-educated statistical workforce capable of explaining Bayesian methods and practical, specific guidelines for prior selection is necessary to make these advanced models more rigorous, systematic, and interpretable.

2.2 Bayesian Model Averaging

While model selection procedures have advanced, particularly with the development of the Akaike information criteria (AIC) (Akaike, 1973) and the Bayesian analog, the BIC, this approach selects incorrect models for BMD estimations nearly as frequently as it identifies correct models (West et al., 2012).

BMA begins to address the challenges of single model selection among parametric models by incorporating uncertainty around the model choice into dose-response assessment.

Model averaging is a technique that conducts inference across a set of models, accounting for uncertainty by estimating the dose-response relationship as a weighted sum of individual models (Wheeler et al., 2020). This general approach (not only in the Bayesian field) demonstrates superior properties to the "best model" approach (Wheeler and Bailer, 2007; Wheeler and Bailer, 2009; Shao and Gift, 2014), and some regulatory agencies now support model averaging in dose-response assessment (EFSA Scientific Committee, 2022). While the U.S. EPA has noted the method's usefulness and has begun to apply model averaging in some cases, the complexity of the modeling and the risk of divergent results have limited a full endorsement (U.S. EPA, 2012).

In traditional Bayesian inference, a single model updates the posterior distribution of the parameters given a prior distribution and the data. BMA extends the concept by also placing prior distributions on the models under consideration and weighting the estimates from the component models by the posterior distribution of the models (Hoeting et al., 1999). By conducting inference both within and across models, BMA reduces the risk of selecting a single incorrect model (Hoeting et al., 1999).

Barnard first introduced the concept of model combination to the statistics literature in 1963. In 1978, Leamer advanced the idea by exploring how the posterior distribution of a quantity of interest could be calculated as a weighted average of the posterior distributions from multiple models, with weights corresponding to each model's posterior probability (Leamer, 1978).

Madigan and Raftery (1994) demonstrated that averaging a subset of possible graphical models provided superior predictive accuracy to a single model, as evaluated by the logarithmic scoring rule. Raftery et al. (1997) then extended this to linear models while also demonstrating the value of a selection mechanism for choosing a smaller subset of possible models before conducting model averaging.

Computing each model's marginal likelihood is essential for BMA. Marginal likelihood represents the probability of the data given the model after integrating out the model parameters. This value is calculated by integrating over the model's parameter space, often approximated by Monte Carlo methods (Raftery et al., 2005), though other methods such as Laplace approximation (Raftery, 1996) have also been applied. Many papers on BMA have also used the BIC, which evaluates models in terms of fit and number of parameters (Schwarz, 1978) or functions thereof, to approximate marginal likelihood (Bailer, Noble, and Wheeler, 2005).

2.2.1 Application to Dose-Response Assessment

There have been many approaches to model averaging for BMD estimation using both frequentist and Bayesian methods (Bailer, Noble, and Wheeler, 2005; Morales et al. 2006; Faes, Aerts, Geys, and Molenberghs, 2007; Wheeler and Bailer, 2007; Wheeler and Bailer, 2009; Shao and Gift, 2014; Simmons et al., 2015; Shao and Shapiro, 2018). In the Bayesian setting, the initial applications of model averaging focused on dichotomous data. Morales et al. (2006) formulated a method to account for model uncertainty in BMD calculations when examining the effects of arsenic exposure on lung cancer. Morales et al. (2006) combined a set of linear models to calculate BMD. To estimate marginal likelihood, they attempted both a Laplace approximation method from Raftery (1996) and an MCMC method from Carlin and Chib (1995). Morales et al. (2006) reported that Laplace approximation was more straightforward and, in specific simulations, accurate. Both methods indicated that model average results differed from the best-fitting model and that while the posterior predictive distributions

peaked at similar locations, the BMA result indicated greater risk at lower doses.

Moving beyond simple linear models, Shao and Small (2011) performed BMA by averaging the Logistic and Quantal-Linear risk functions. In 2012, Shao explored further methods in BMA by combining Logistic, Quantal-Linear, and Probit models, using three forms of historical information (a pooled data approach, a Bayesian hierarchical model, and a power prior). However, the results did not differ in the BMD estimation.

West et al. (2012) found that the models with an underlying Logistic risk function were misspecified when fit by a Quantal-Quadratic risk model, leading to overestimated BMD estimates and risk calculations. This highlights the importance of considering model uncertainty in dose-response estimation.

Guha et al. (2013) compared BMA with a non-parametric estimation method for evaluating BMD, finding that the BMA was superior to choosing a single best model.

Shao and Gift (2014) also conducted BMA for BMD estimation, also using two methods for estimating the marginal likelihood: BIC and an MCMC-based density estimation. They found that the two approaches for estimating marginal likelihood produced similar results in the limited datasets they explored. However, the BIC method assigned relatively higher posterior weights on similar models given the penalty for model parameters. Conversely, the MCMC method was more flexible, but specifying (or eliciting) priors was a challenging and subjective task, compounded by how sensitive the posteriors were to the priors (Shao and Gift, 2014). This study was also one of the first to develop an approach for estimating BMD and BMDL estimates for continuous dose-response data. Their results indicated that the BMA BMD estimates were more reliable and had a smaller bias than those of the best-fitting model (Shao and Gift, 2014).

Fang et al. (2015) applied Bayesian model averaging to their eight reparametrized parametric dose-response models. They used a uniform prior over the models and calculated marginal likelihood using a geometric bridge sampler (Fang et al., 2015) developed by Meng and Wong (1996). Simmons et al. 2015 also developed reparametrized models for BMD estimation from Logistic, Probit, Quantal-Linear, and Quantal-Quadratic models using a non-parametric density estimation approach for approximating the marginal likelihood of the model from Rosenblatt (1956).

Despite the appeal of BMA, the choice of priors on the models, the evaluation of model probabilities, and computation demands complicate the method. Regarding the selection of model priors, there have been limited attempts to go beyond the uniform priors. The EFSA Scientific Committee (2022) suggests uniform priors unless one has a good reason to do otherwise, though no method for capturing this belief is proposed. Shao and Gift (2014) noted that this assumption is reasonable when considering a set of historically-effective models when assessors lack specific functional knowledge of the agent under consideration.

However, some parametric models have properties consistent with certain classes of toxins or routes of exposure, suggesting that it may be possible to place priors on models if we can quantify knowledge of the biological process.

Furthermore, parametric model averaging for BMD estimation involves the choice of parametric models by the modeler, and performance often depends on the models chosen for inclusion. Wheeler and Bailer (2007) showed that results vary based on the number and type of parametric models included in the averaged results. In practice, these results may be minimal (Shao and Gift, 2014).

This issue is compounded, as many models can degenerate into the same model during fitting. For example, bounds on parameters can cause models like the Weibull and Multistage models to degenerate into the Quantal-Linear model, biasing results and complicating model weight construction and inference (Wheeler et al., 2020).

BMA methods are often computationally demanding, making them less appealing for larger datasets. Attempts to reduce computational demand typically focus on the calculation of the marginal likelihood or on the reduction of the model space. For example, Wheeler et al. (2020) used a Laplace approximation to estimate marginal likelihood. They also applied priors on the models under consideration that ensured more complicated models that degenerated to simpler models (already included in the model averaging) would have weights close to zero, thereby preventing the complicated models from being counted twice. This method increases the speed of the analysis, allows for better reproducibility, and produces superior results to the single "best" model in dose-response assessment (Wheeler et al., 2020).

BMA can provide a more robust method for estimating dose-response functions than choosing a single best-fitting model by addressing uncertainty in model prediction and reducing the risk of overfitting the model, particularly in cases where model choice is not straightforward. However, challenges persist, such as model class inclusion, the estimation of marginal likelihoods, prior selection over the model space, and computational demand.

2.3 Gaussian Process Regression

GPR is a Bayesian non-parametric method that can capture the relationship between inputs and outputs using a theoretically infinite number of parameters (Williams, 1998). It is particularly useful when the relationship between variables is non-linear (Rasmussen and Williams, 2006).

If we consider f as an unknown function that maps inputs x to outputs y, we can attempt to model f using Bayesian linear regression, other parametric Bayesian means, or model averaging approaches. However, mapping input variables into a new feature space by transforming them or exploring a limited set of potential parametric models is not exhaustive in the search for the target function. A GPR method allows the data to effectively set the complexity of the estimated function (Schulz, Speekenbrink, and Krause, 2018).

A Gaussian process (GP) is a collection of random variables (RVs) where any set of such RVs have a multivariate Gaussian distribution (Schulz, Speekenbrink, and Krause, 2018). In GPR, outputs are modeled as:

$$y_t = f(x_t) + \epsilon$$

where both $f(x)$ and ϵ are treated as random variables (RVs). The uncertainty around $f(x)$ diminishes with additional observations, while ϵ reflects natural randomness (Schulz, Speekenbrink, and Krause, 2018).

Assuming that $f(x)$ is distributed as a Gaussian Process (GP), we can denote:

$$f(x) \sim \text{GP}\left(m(x), k(x, x')\right)$$

A GP is fully parameterized by its mean function $m(x)$ and a covariance function $k(x, x')$ (Rasmussen and Williams, 2006), commonly called a kernel.

The mean function represents the expected value of the function at x (i.e., the average of all the possible functions evaluated at that input) and is typically set to zero. The kernel function $k(x, x')$ generates a covariance matrix between different inputs. We typically assume that points farther from each other are less correlated than closer points and behave less similarly (Schulz, Speekenbrink, and Krause, 2018).

A recent review of BMD methodologies found that non-parametric models (including GPR) were able to fit dose-response patterns more flexibly than traditional parametric approaches; however, such methods were also more likely to overfit the data (Jensen, Kluxen, and Ritz, 2019). Jensen, Kluxen, and Ritz (2019) also noted that these models typically yield conservative BMDL estimates.

2.3.1 Monotinicty Constraints

While one of the benefits of GPR is the flexibility in modeling without assumptions of parametric forms, it is sometimes necessary or reasonable to incorporate domain knowledge about a process by adding constraints. Imposing a constraint on a GP because of a belief about the underlying biological process is a form of "physics-constrained machine learning" (Swiler et al., 2020). Bound and shape constraints (Riihimäki and Vehtari, 2010) are among the methods for incorporating domain knowledge into the GPR.

Bound constraints limit the region of interest for the function. For example, as biological measures are typically positive (EPA, 2012), our techniques should be constrained to positive outcomes. Additionally, we can limit the outcome space to values between 0 and 1, as our outcomes of interest are probabilities.

Shape constraints ensure that the function adheres to specific structural properties, such as convexity or monotonicity (Riihimäki and Vehtari, 2010). A monotonic function is either non-decreasing or non-increasing over its domain. Dose-response curves for toxins are typically assumed to be non-decreasing monotonic functions (EPA, 2012). Such monotonicity constraints are typically applied by enforcing the derivative of the function to be nonnegative ($f' \geq 0$).

Several methods have been proposed for incorporating monotonicity. For example, Riihimäki and Vehtari (2010) enforced monotonicity by constraining the likelihood of the derivative observations, while Lopez-Lopera et al. (2018) used splines subject to nonnegativity constraints on their knots, and Swiler et al. (2020) did so by ensuring convexity through nonnegativity constraints to higher-order derivatives of f. Additionally, Lin and Dunson (2014) developed two methods for Bayesian monotone regression using GP projections: sampling from an unconstrained GP prior or posterior onto the space of a monotone function, which ensured that the final model was monotonic. Riihimäki and Vehtari (2010) showed that a GP could be trained on derivative observations, forcing the monotonicity constraint on the final model. Additionally, Guha et al. (2013) used an integrated exponential of the latent function to ensure monotonicity across the domain.

One of the major challenges of GP application is that of scaling, as inferring a posterior distribution requires matrix inversion, which scales cubically $O(n^3)$ as observations N increase (Rasmussen and Williams, 2006). However, there have been several approximations to reduce this burden, such as Greedy approximations, reduced-rank approximations of the Gram matrix (Rasmussen and Williams, 2006), or using a nearest neighbor approach to group observations, minimizing the number of computations (Ren et al., 2023; Khoshnevis, Ren, and Braun, 2024). Given the limited data from toxicology

studies – even while incorporating multiple species, outcomes, and both sexes – approaches that seek to evaluate toxicology experiments should not require more computation than a modern computer could provide. As the data grows or researchers explore epidemiological data, approximation methods that reduce computation demand will become more important.

2.3.2 Application to Dose-Response Assessment

Even as GPs have been applied to medicine and dosimetry, particularly in radiation exposure (Russu et al., 2011; Kim et al. 2020; Siivola, Weber, Vehtari, 2021; Balanya et al. 2022), the application of GPR to dose-response modeling in human health risk assessment is still nascent. Shi et al. (2012) developed a semiparametric model that combined a parametric mixed-effects model with a GPR model to produce a multilevel model analysis that could be applied to medical research, as curves in medicine often included grouped data with multiple observations coming from the same individual either over time or at different drug dose levels. To expand on the functional forms that could be practically explored with a purely parametric set of models, Shi et al. (2012) included a GPR (rather than a local polynomial model or spine smoothing model, as discussed in Kelly and Rice (1990)) because this model led to a more consistent dose-response curve prediction. However, the goal was not to develop a population dose-response curve, but rather to develop individualized dose-response curves for each patient taking a given drug to support more individualized patient treatment regimens (Shi et al., 2012). The use of the non-parametric model within a fixed effects model merits further attention as we incorporate additional data sources into our future studies.

Guha et al. (2013) developed an approximation of a GP using an integrated exponential random walk (Brownian motion) process and found it offered a more accurate approximation of the dose-response curve and BMD estimate than the BMA approach they also attempted.

More recently, a research team at Harvard developed an R package for estimating a causal exposure-response function (CERF), with the associated uncertainty estimates, for continuous exposures using a GP (Khoshnevis, Ren, and Braun, 2024). The program, `GPCERF`, first uses the data to set the optimal hyperparameters to achieve covariate balance (in support of causal inference). These hyperparameters were used to generate a pseudo-population that was a weighted version of the original data so that the different exposure groups had balanced covariates, supporting the causal inference goal of the program (Khoshnevis, Ren, and Braun, 2024). It then estimated the curve based on the dataset using a standard GP and a nearest-neighbor GP (nnGP) (Khoshnevis, Ren, and Braun, 2024). The nnGP was an attempt to reduce the computation cost of the program since it was designed to evaluate large amounts of observational data (rather than the more limited experimental data that we examine).

Two of the researchers who developed the program, alongside a group of additional researchers, also applied this `GPCERF` program to an environmental epidemiology problem: the effect of PM2.5, NO_2, and O_3 on all-cause mortality among a large population of Medicare enrollees in the U.S. (using the nnGP form) (Ren et al., 2023). Ren et al. (2023) demonstrated the usefulness of the `GPCERF` package, particularly for large datasets, and showed they can detect change points (similar to PoDs) in their analysis. Interestingly, Ren et al. 2023 did not enforce a monotonicity constraint on their GP, and the posterior reveals non-monotonic curves for NO_2 and O_3. These are noteworthy trends in the data, but we maintain our belief in the need for monotonicity constraints in our approach.

3 Methods

3.1 Data Description

In this paper, we explore two datasets: a simulation set and an experimental antimony set.

3.1.1 Simulation Data

For the simulation set, we selected a set of four doses similar in spacing and range (0.00, 0.50, 2.00, and 6.00 mg/kg-bw/day) to our antimony dataset and generated probabilities of a response from a Log-Probit model with known parameters (where $\beta_0 = 0.5$, $\beta_1 = 1.5$, and $\gamma_0 = 0.1$) and 50 subjects in each dose group. We chose the Log-Probit model because it is commonly used in toxicology modeling and demonstrates a sigmoidal curve on a logarithmic scale. It also assumes that there is a dose at which there is no response, which is consistent with our beliefs for an experiment that includes a zero dose level. Furthermore, the model is not the simplest nor the most complex among our nine models under consideration. However, other models may also have worked equally well or better, and the choice of the Log-Probit model and the parameters we selected for our "true" model here are not necessarily superior to other models or parameters. The goal was to determine the conditions under which the parametric models would be able to recover a dose-response relationship when the true data came from a biologically-plausible dose-response curve. We acknowledge that since the data were generated from one of the parametric models we will test, this may influence our results. However, evaluating the Bayesian parametric, BMA, and GPR models that result from fitting this data may still prove valuable to understanding how Bayesian methods perform in recovering known dose-response relationships.

After simulating the probability of a response, we applied the binomial likelihood to estimate the number of outcomes that we will call our "true" cases for the simulation set. With these true case counts, we injected noise in the outcomes at each dose level. The addition of noise simulates variability in cancer development among individual subjects and the stochastic nature of carcinogenesis (Barrett, 1993). We modeled the noise as coming from a discrete uniform (-3, 3) distribution while ensuring that the final count was nonnegative and did not surpass the number of animals in the dose group. This method is imperfect, but we did not find a superior method to simulate random noise in cancer outcomes.

3.1.2 Experimental Antimony Data

For our experimental set, we analyze the results of a two-year toxicology experiment by the National Toxicology Program (NTP). In the study, animal subjects were exposed to antimony trioxide via inhalation for six hours per day, six days per week, for two years. The experiment applied the same procedure to B6C3F1/N mice and Wistar Han rats. There were originally 60 male and 60 female rodents (for each species) in the four exposure groups (0, 3, 10, and 30 mg/m^3). Ten animals were sacrificed at week 53 for evaluation, and those are not included in our final analysis.

Here, we focus on the group of female B6C3F1/N mice and the outcome of alveolar/bronchiolar adenoma or carcinoma. In the study, female mice were the most sensitive group, and such choice is the typical approach to modeling risk when human studies are unavailable or are of insufficient quality.

We chose to explore the risk function for alveolar/bronchiolar adenoma or carcinoma rather than another form of cancer (or total cancer risk) as these are cancers of the respiratory system, consistent with expectations from inhalation exposure and of direct interest to human health. Furthermore, this particular set appears to be more challenging to model than the other forms of cancer observed in the mice, as the responses are not increasing at all levels. The incidence of cancer in the highest-exposure group is lower than the incidence in dose group three.

As previously mentioned, it is standard to convert exposure concentrations to exposure doses in terms of average body weight. We follow Schildroth et al.'s (2018) procedures, with concentrations converted to the following doses: 0.00, 0.63, 2.12, and 6.35 mg/kg-bw/day.

The outcomes of the two-year toxicology study present some challenges to explanation and modeling. Survival in the two highest concentration groups was significantly lower than in the control groups (NTP, 2017). Among female mice, these deaths were primarily attributed to malignant lymphoma and lung inflammation (NTP, 2017) before they might have otherwise developed an adenoma or carcinoma. Table 1 shows the doses and incidence counts we use in this paper. The NTP attempted a Poly3 correction for early deaths (to provide an updated tumor count). Even with this correction, the incidence among the final group remains low. This presents a challenge frequently seen in real-world data, and additional solutions are needed to address this form of data.

Table 1: Incidence of Alveolar/Bronchiolar Adenomas and Carcinomas in Female Mice by Dose Group After Poly3 Correction

Dose group (mg/kg-bw/day)	**Incident tumors**	**Total animals***
0	3	47
0.63	22	46
2.12	27	44
6.35	19	44

*Number of subjects alive at the time of first occurrence of tumor.
Adapted from Schildroth et al. (2018).

The NTP concluded that there was "clear evidence of carcinogenic activity" based on the increases in alveolar/bronchiolar adenoma and carcinoma of the lung female B6C3F1/N mice and recommended that the chemical be labeled as "reasonably anticipated to be a human carcinogen" given the evidence in the animal and mechanistic studies.

We note that we do not have a known, true function for the antimony dose-response curve against which we can compare our estimates.

Given the results of the NTP findings, the lack of a full risk assessment by any regulatory body, the importance and prevalence of antimony trioxide in industry, the challenge of this dataset, our understanding of the mechanism of action, and the potential health burden of antimony trioxide, a dose-response assessment is warranted to explore the health impact of this chemical on the population and among high-exposure sub-populations. As such, we chose to apply the methodology developed in this paper to estimate a set of dose-response curves to antimony trioxide from the NTP's experiment.

3.2 Bayesian Parametric Methods

3.2.1 Implementation

We employed Bayesian parametric modeling to fit nine biologically-plausible models. In Table 2, we show the parameterization, priors, and constraints for each model. In these equations, β_0 is typically the intercept parameter, β_1 is the slope, and γ_0 is the baseline risk parameter (representing the proportion of tumor incidence at dose zero). The exceptions to these standard interpretations are for the Multistage model (two-stage) where β_1 is the linear term and β_2 is the quadratic term; the Weibull model, where β_0 is the scale parameter and β_1 is the shape parameter; and the Dichotomous Hill model where κ is the Hill coefficient, γ_0 is the lower asymptote parameter (similar to the intercept in other models), and θ_0 is the dose-response parameter representing the dose at which the tumor incidence is halfway between the baseline and the maximum response.

The EPA recommends eight of these models (Logistic, Probit, Quantal-Linear, Multistage (two-stage), Log-Logistic, Log-Probit, Weibull, and Dichotomous Hill) for benchmark dose estimation given their biological plausibility in modeling dichotomous outcomes for toxicology studies, and we also included the Quantal-Quadratic model, which researchers have included in seminal Bayesian dose-response assessments (Fang, Piegorsch, and Barnes, 2015; Shao and Shapiro, 2018). For these models, θ is the vector that includes the parameters of the model, with either two or three parameters in each of the nine models. In line with previous work, we assume that the unknown parameters are independent (Shao and Small, 2011; Shao, 2012; Fang, Piegorsch, and Barnes, 2015), such that $\pi(\theta) = \pi(\theta_1, \theta_2, \theta_3) = \pi(\theta_1)\pi(\theta_2)\pi(\theta_3)$ in the three-parameter case (and the two-parameter case, similarly).

We use a set of semi-informative priors from the 2022 U.S. EPA's BMDS guidance for dichotomous models (U.S. EPA, 2022). In the parameterizations we present, we avoid the need for a Logistic transformation of the priors to the [0,1] bound (suggested by the U.S. EPA (2022)) by using a Beta (1,1) prior for those constrained parameters. We also adapt and update the constraints of Fang et al. (2015) on the model, ensuring that they are consistent with biological expectations, such as a positive, non-infinite slope (as in the Weibull model).

For all parametric models, we estimate the probability of a response $R(d)$ at a given dose level d. The likelihood of a given number of outcomes Y_i is distributed as a binomial distribution with N_i number of test subjects and probability $R(d)$, as follows:

$$Y_i \sim \text{Bin}(N_i, R(d)))$$

We coded this project in Stan and R. For each model simulation, we had four MCMC chains with 5,000 burn-in values and 20,000 iterations, and we thinned every other sample to reduce autocorrelation. We used a No-U-Turn-Sampler to sample from the posterior. Regarding notation, our priors follow a Normal (N), lognormal (LN), or Beta distribution, and Φ is the cumulative distribution function of the standard Normal distribution.

Table 2: Parametric Model Descriptions

Model	Equation	Priors	Constraints
Logistic	$R(d) = \text{logit}^{-1}(\beta_0 + \beta_1 \cdot d)$	$\beta_0 \sim \mathcal{N}(0, 2)$ $\beta_1 \sim \mathcal{LN}(0, 2)$	$\beta_0 \in \mathbb{R}$ $\beta_1 \geq 0$
Probit	$R(d) = \Phi(\beta_0 + \beta_1 \cdot d)$	$\beta_0 \sim \mathcal{N}(0, 2)$ $\beta_1 \sim \mathcal{LN}(0, 1)$	$\beta_0 \in \mathbb{R}$ $\beta_1 \geq 0$
Quantal-Linear	$R(d) = 1 - \exp\left(-(\beta_0 + \beta_1 \cdot d)\right)$	$\beta_0 \sim \mathcal{N}(0, 2)$ $\beta_1 \sim \mathcal{LN}(0, 1)$	$\beta_0 \geq 0$ $\beta_1 \geq 0$
Quantal-Quadratic	$R(d) = \gamma_0 + (1 - \gamma_0)\left(1 - \exp(-\beta_1 \cdot d^2)\right)$	$\gamma_0 \sim \text{Beta}(1, 1)$ $\beta_1 \sim \mathcal{LN}(0, 1)$	$0 \leq \gamma_0 < 1$ $\beta_1 \geq 0$
Multistage (two-stage)	$R(d) = 1 - \exp\left(-(\beta_0 + \beta_1 \cdot d + \beta_2 \cdot d^2)\right)$	$\beta_0 \sim \mathcal{LN}(0, 1)$ $\beta_1 \sim \mathcal{LN}(0, 1)$ $\beta_2 \sim \mathcal{LN}(0, 1)$	$\beta_0 \geq 0$ $\beta_1 \geq 0$ $\beta_2 \geq 0$
Log-Logistic	$R(d) = \gamma_0 + (1 - \gamma_0)\left(\frac{1}{1+\exp(-\beta_0 - \beta_1 \cdot \log(d))}\right)$	$\gamma_0 \sim \text{Beta}(1, 1)$ $\beta_0 \sim \mathcal{N}(0, 10)$ $\beta_1 \sim \mathcal{LN}(0, 1)$	$0 \leq \gamma_0 < 1$ $\beta_0 \in \mathbb{R}$ $\beta_1 \geq 0$
Log-Probit	$R(d) = \gamma_0 + (1 - \gamma_0)\Phi(\beta_0 + \beta_1 \cdot \log(d))$	$\gamma_0 \sim \text{Beta}(1, 1)$ $\beta_0 \sim \mathcal{N}(0, 10)$ $\beta_1 \sim \mathcal{LN}(0, 1)$	$0 \leq \gamma_0 < 1$ $\beta_0 \in \mathbb{R}$ $\beta_1 \geq 0$
Weibull	$R(d) = \gamma_0 + (1 - \gamma_0)\left(1 - \exp\left(-\exp(\beta_0) \cdot d^{\beta_1}\right)\right)$	$\gamma_0 \sim \text{Beta}(1, 1)$ $\beta_0 \sim \mathcal{N}(0, 10)$ $\beta_1 \sim \mathcal{LN}(1.5, 1)$	$0 \leq \gamma_0 < 1$ $\beta_0 \in \mathbb{R}$ $\beta_1 \geq 1$
Dichotomous Hill	$R(d) = \gamma_0 + (1 - \gamma_0)\left(\frac{d^k}{d^k + \theta_0^k}\right)$	$\gamma_0 \sim \text{Beta}(1, 1)$ $\theta_0 \sim \mathcal{N}(-3, 3.3)$ $k \sim \mathcal{LN}(\ln(2), 0.5)$	$0 \leq \gamma_0 < 1$ $\theta_0 \geq 0$ $\kappa \geq 0$

After fitting each model, we extracted the posterior distributions of the parameters and generated posterior predictive distributions to model the dose-response curve f. We extracted the mean estimates at the dose levels with a 95% credible interval around each. To visualize the dose-response curve, we then plotted the resulting mean predicted response at each dose level, a 95% credible interval around the estimates, and the observed (or simulated) data points.

3.3 Bayesian Model Average Methods

Once we extracted the posterior samples from the nine parametric models, we continued to the model averaging component of our methods. Though model averaging over an extensive model space may offer superior results, this exhaustive exploration and summation is often too computationally demanding to be practical (Raftery et al., 1997). Further, such an approach may be unnecessary when there are a limited number of parsimonious, data-supported models given by some selection mechanism (Madigan and Raftery, 1994). We applied the EPA's recommendations and past research to limit our model exploration using the nine models supported for dichotomous toxicology data. We also gain confidence in including these limited models as Draper (1995) discussed the importance of averaging models with varying link functions and different error structures to support robust model averaging results.

3.3.1 Implementation

Building upon our parametric models, we consider the dose-response function f. The posterior distribution of f given our data, Z, is given by,

$$p(f|Z) = \sum_{k=1}^{K} p(f|Z, M_k)p(M_k|Z)$$

where $p(f|Z, M_k)$ is the posterior predictive distribution of f given data Z under model M_k and $p(M_k|Z)$ is the posterior probability of model M_k given data Z.

This gives us the average of the posterior predictive distribution of our dose-response function under each of the models under consideration, weighted by their corresponding posterior probability.

To determine the posterior probability of each model $p(M_k|Z)$, we use Bayes' rule,

$$p(M_k|Z) = \frac{p(Z|M_k)p(M_k)}{\sum_{l=1}^{K} p(Z|M_l)p(M_l)}$$

where $p(Z|M_k)$ is the marginal likelihood of the data under model M_k and $p(M_k)$ is the prior probability of model M_k. We place uniform priors on our models until we design a better way to quantify this knowledge.

To calculate the posterior probabilities of each model, we need to calculate $p(Z|M_k)$, and this is one of the challenges of BMA because $p(Z|M_k)$ is given by the integrated likelihood of M_k considering the parameters θ_k of M_k and the prior density of the parameters under that model, conditional on the set of models under consideration (here, our nine parametric models).

$$p(Z \mid M_k) = \int \cdots \int p(Z \mid \theta_k, M_k)p(\theta_k \mid M_k)\, d\theta_k$$

With estimates from the posterior predictive distributions of our nine component models and the weights for each model, we then produce our dose-response function f over the range of doses, as previously described.

3.3.2 Marginal Likelihood Estimation Using Bridge Sampling

To calculate the marginal likelihood of our individual models $p(Z|M_k)$, we employ bridge sampling, a Monte Carlo method that incorporates elements of accept-reject sampling. Bridge sampling can reduce simulation errors by an order of magnitude compared to importance sampling with lower computation demand than MCMC methods (Meng and Wong, 1996). This is important as we seek accurate estimates that can scale, particularly with larger datasets in our future research for estimating dose-response curves across multiple studies, species, sexes, and outcomes.

We will use the premade `bridgesampling` package, which can be accessed from (CRAN, 2022; Gronau et al., 2022). (We will develop our own code for this approximation in our future work.) Though initially used to estimate the Bayes factor in model comparison (Diciccio et al., 1997), bridge sampling has expanded into an accurate and efficient method for estimating marginal likelihood (Frühwirth-Schnatter, 2004).

Bridge sampling requires samples from the posterior distribution of our component models (Gronau et al., 2017), which we have already calculated in our estimation of the parametric models.

The program that we use is based on the identity that,

$$1 = \frac{\int p(Z|\theta_k, M_k)\, p(\theta_k|M_k)\, h(\theta)\, g(\theta)\, \mathrm{d}\theta}{\int p(Z|\theta_k, M_k)\, p(\theta_k|M_k)\, h(\theta)\, g(\theta)\, \mathrm{d}\theta}$$

where $p(Z|\theta_k, M_k)$ is the likelihood of the observed data given the parameters, $p(\theta_k|M_k)$ is probability of the parameters in a given model, $g(\theta)$ is the proposal distribution, and $h(\theta)$ is the bridge function. After multiplying both sides of the identity by the marginal likelihood and rearranging, we have that,

$$p(Z|M_k) = \frac{\int p(Z|\theta_k, M_k)\, p(\theta_k|M_k)\, h(\theta)\, g(\theta)\, d\theta}{\int \frac{p(Z|\theta_k, M_k)\, p(\theta_k|M_k)}{p(Z|M_k)}\, h(\theta)\, g(\theta)\, d\theta} = \frac{\int p(Z|\theta_k, M_k)\, p(\theta_k|M_k)\, h(\theta)\, g(\theta)\, d\theta}{\int h(\theta)\, g(\theta)\, p(\theta_k \mid Z, M_k)\, d\theta}$$

This equals the following,

$$p(Z|M_k) = \frac{\mathbb{E}_{g(\theta)}\left(p(Z|\theta_k, M_k)\, p(\theta_k|M_k)\, h(\theta)\right)}{\mathbb{E}_{\text{post}}\left(h(\theta)\, g(\theta)\right)}$$

and we can see that the marginal likelihood can be approximated by,

$$\hat{p}(Z|M_k) = \frac{\frac{1}{N_2}\sum_{i=1}^{N_2} p(y \mid \tilde{\theta}_i)\, p(\tilde{\theta}_i)\, h(\tilde{\theta}_i)}{\frac{1}{N_1}\sum_{j=1}^{N_1} h(\theta_j^*)\, g(\theta_j^*)}$$

where $\tilde{\theta}_i \sim g(\theta)$ are samples from the proposal distribution and $\theta_j^* \sim p(\theta_k \mid Z, M_k)$ are samples from the posterior distribution. As such, we need samples from the proposal and posterior distributions to estimate the marginal likelihood.

We leave the tuning of the bridge and proposal distributions to the `bridgesampling` program for now. As with accept-reject sampling, it is important that the proposal distribution is similar to the posterior. A Normal distribution with hyperparameters set based on the posterior is the standard choice (Overstall and Forester 2010). The program we employ sets a Multivariate Normal (*MVN*) proposal distribution with mean equal to the mean vector of samples and the covariance matrix as the covariance matrix of the posterior samples (Gronau et al., 2022).

The program also sets the bridge function. The `bridgesampling` package uses the optimal bridge function expressed by Meng and Wong (1996), who showed that this formulation minimizes the relative mean squared error and is given by,

$$h(\theta) = C \cdot \frac{1}{s_1 p(Z|\theta_k, M_k) p(\theta_k|M_k) + s_2 p(Z|M_k) g(\theta)}$$

with $s_1 = \frac{N_1}{N_2+N_1}, \quad s_2 = \frac{N_2}{N_2+N_1}$, and a constant C. This function depends on the value we are trying to estimate. To address this problem, the program uses an iterative method to update estimates from an initial guess at the marginal likelihood until it has converged.

The final result is an estimate of the marginal likelihood for each model, which we then use to estimate the posterior probability of our models.

3.4 Gaussian Process Regression Methods

3.4.1 Implementation

For our GPR method, we assume that $f(x)$ is distributed as a GP:

$$f(x) \sim \text{GP}\left(m(x), k(x, x')\right)$$

As is standard practice, we set our mean function $m(x)$ equal to zero. The RBF kernel can encode the belief that more distant points are less related to one another. It is defined by,

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\lambda^2}\right)$$

This kernel includes two hyperparameters: λ, which represents the length scale, and σ_f^2, which is the signal variance. These can be updated to influence the correlation between points and the final form of the function (Schulz et al., 2016).

Given our data $Z = \{\mathbf{X}_t, \mathbf{y}_t\}$, our goal was to predict the function f at new input points $\mathbf{X}_*$ by drawing f from the posterior $p(f \mid Z)$. We know that our previous observations $\mathbf{y}_t$ and function values $\mathbf{f}$ follow a *MVN* distribution:

$$\begin{bmatrix} \mathbf{y}_t \\ \mathbf{f} \end{bmatrix} \sim \mathit{MVN}\left(\mathbf{0}, \begin{bmatrix} \mathbf{K}(\mathbf{X}_t, \mathbf{X}_t) + \sigma_\epsilon^2 \mathbf{I} & \mathbf{K}(\mathbf{X}_t, \mathbf{X}_\star) \\ \mathbf{K}(\mathbf{X}_\star, \mathbf{X}_t) & \mathbf{K}(\mathbf{X}_\star, \mathbf{X}_\star) \end{bmatrix}\right)$$

Where $\mathbf{I}$ is the identity matrix and σ_ϵ^2 is the variance of epsilon (or the assumed noise level of the observations). The conditional distribution of $p(\mathbf{f} \mid \mathbf{X}_t, \mathbf{y}_t, \mathbf{X}_\star)$ is a GP with mean:

$$m_t(\mathbf{x}) = \mathbf{K}(\mathbf{x}, \mathbf{X}_t)\left[\mathbf{K}(\mathbf{X}_t, \mathbf{X}_t) + \sigma_\epsilon^2 \mathbf{I}\right]^{-1} \mathbf{y}_t$$

and covariance

$$k_t(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}') - \mathbf{K}(\mathbf{x}, \mathbf{X}_t) \left[\mathbf{K}(\mathbf{X}_t, \mathbf{X}_t) + \sigma_\epsilon^2 \mathbf{I}\right]^{-1} \mathbf{K}(\mathbf{X}_t, \mathbf{x}').$$

Though one of the major appeals of using a GP is the lack of constraints or parametric form imposed on the model, we want to ensure that the functional form is consistent with our understanding of a physical process. As noted, toxicology risk functions typically demonstrate a monotonic, nondecreasing relationship with dose. To enforce monotonicity, we use the sum of exponentials (in a function we will call g) of our latent function f_i at different dose levels, which ensures that the function values are non-decreasing.

$$g_i = \sum_{i=1}^{i} \exp(f_i)$$

Researchers have explored this method to ensure a nondecreasing function through a Brownian motion process (Guha et al., 2013), as well as in binary regression (Wood and Kohn, 1998).

We also employ a bound constraint using the Logistic function to map our values onto the [0,1] range, ensuring that GP output can be interpreted as probabilities.

$$R(d) = \frac{g_i}{1 + g_i}$$

These probabilities at each dose level are then used in our binomial likelihood, as our expected observations of cases are modeled by a binomial likelihood based on the number of subjects N_i at each dose level d and the probability of developing cancer $R(d)$.

$$Y_i \sim \text{Bin}(N_i, R(d))$$

where Y_i are realizations of the latent function f.

We coded the model in Stan and ran the code in R. We sampled from the posterior distribution using 20,000 iterations, with 5,000 burn-in values, thinning every other sample. We extracted the mean estimate at each dose level and calculated the 95% credible intervals for each predicted probability at the dose levels. We plotted these estimates against the observed data.

3.4.2 Optimizing Hyperparameters

In certain instances, it is possible to infer GP hyperparameters from the data, but this is not common (Schulz et al., 2018). Instead, researchers typically select point estimates for these values by choosing hyperparameters that maximize the marginal likelihood of the model (Williams and Rasmusen, 2006).

We attempted to elicit our hyperparameters through inference from the data, but this only recovered the prior. We believe this is because of how limited our data was. We then fit our model with a range of hyperparameter pairs and optimized these to elicit the values that maximized the marginal likelihood of the model given our data.

The marginal likelihood of the observed data $\mathbf{y} = [y_1, y_2, \ldots, y_n]$, given the hyperparameters, integrates out the latent function values f:

$$p(y \mid \theta) = \int p(y \mid f) p(f \mid \theta) \, \mathrm{df},$$

where $\theta = (\sigma_f^2, \lambda)$ are the hyperparameters.

We searched for the hyperparameters that maximized the marginal likelihood $p(y \mid \theta)$, which involved solving for,

$$\theta^* = \arg\max_{\theta} p(y \mid \theta)$$

We then extracted these optimal values and applied them to our GPR.

3.5 Convergence diagnostics

For the Bayesian parametric models and the GPR model, we evaluated our MCMC samples using Rhat values to check for convergence, trace plots (for adequate mixing of the chains), autocorrelation plots (to evaluate autocorrelation within chains), and pairs plots (to check for dependencies between pairs of parameters). We also checked that our n_{eff} values showed adequate numbers of independent samples for inference.

3.6 Evaluation Criteria

3.6.1 BIC

Following a standard approach for selecting a single best parametric model, we calculated the BIC value for each model (using the same set of noisy data) and chose the model with the lowest BIC value as our best-fitting model. For each model, we extracted the log-likelihood values from the fitted Stan object and found the maximum likelihood value from the samples as,

$$\hat{L} = \max(\text{log_lik}_1, \text{log_lik}_2, \ldots, \text{log_lik}_N)$$

We then identified the number of parameters p in each model and the number of observations n and calculated the BIC as,

$$\text{BIC} = p\log(n) - 2\hat{L}$$

Though an estimate, a lower BIC value indicates a better fit, accounting for the number of parameters (Chakrabarti and Ghosh, 2010).

3.6.2 Model Predictive Performance: Simulation

For the simulation component (not the antimony dataset, where we did not have access to the true underlying model), we evaluate each model's predictive performance by assessing mean squared error (MSE). For each of the models under consideration, we conducted 10 simulations, adding uniform (-3, 3) noise to the "true" tumor counts (with the constraints noted in the random noise section). For each of these simulations, the Stan model was fit to the noisy data to obtain the posterior distribution of the parameters. We then estimated the expected tumor proportions using these parameters.

We calculated MSE as follows:

$$\text{MSE} = \frac{1}{S}\sum_{i=1}^{S}\left(R(d)^{\text{true}} - R(d)^{\text{est.}}\right)^2$$

Where S is the number of simulations, $R(d)^{\text{true}}$ is the true proportion of tumors at dose level d, and $R(d)^{\text{est.}}$ is the estimated proportion derived from the model under consideration. In doing so, we attempt to estimate model accuracy.

We compared the MSE values across all parametric, BMA, and GPR models in the simulation set to assess predictive performance.

3.6.3 Fit and Credible Intervals

Finally, we compared the estimates provided by the models (along with the 95% credible intervals) with the "true" values in our simulated dataset and the observed values in the antimony dataset. We did this to understand how the models capture or fail to capture the underlying trends of the data and each model's ability to capture observations in the credible intervals.

4 Results

4.1 Simulation Results

4.1.1 Bayesian Parametric Results

Our simulation results show that when a limited amount of data is derived from a known functional form with moderate noise added to each output value, we are able to fit most models relatively well. While some models fit better than others, all the models in this scenario are viable choices for modeling the risk function. Figure 1 shows plots of all our parametric models fitted to the same set of noisy data. As standard practice for selecting a model when limited (or no) mechanistic information in-

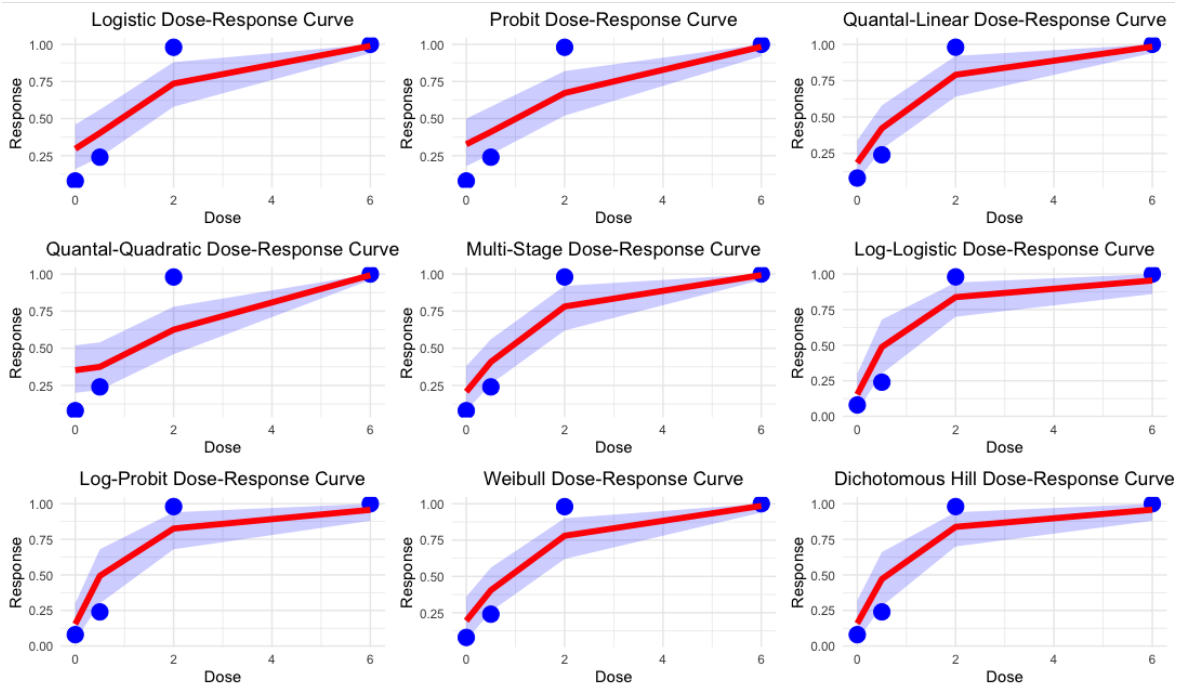


Figure 1: Simulation: Bayesian Parametric Dose-Response Curves

forms model selection, we use BIC to select the "best" dose-response curve. Following this protocol, we select the Dichotomous Hill model. Table 3 presents the BIC values for each of the parametric models. As can be seen in Figure 1 and Table 3, the Log-Logistic model is also a strong contender, with a nearly-identical visual fit and BIC value.

Table 3: Simulation: Parametric Model Comparison

Model	Marginal Likelihood	Posterior Probability	BIC
Logistic	-80.332	0.326	53.326
Probit	-93.783	0.213	65.134
Quantal-Linear	-66.103	0.005	73.154
Quantal-Quadratic	-59.620	0.129	40.156
Multistage (Two-stage)	-88.920	0.024	47.478
Log-Logistic	-64.606	0.097	30.530
Log-Probit	-77.538	0.092	34.369
Weibull	-90.116	0.099	42.902
Dichotomous Hill	-57.147	0.016	30.490

Considering the MSE values over multiple iterations, the Dichotomous Hill, Logistic, and Multistage models offered realizations closest to the "true" formulation (Table 4). These results generally agree with our BIC values, with the Dichotomous Hill again best fitting the data. However, it is important to note that we calculated the BIC values for these models from a single iteration of curve fitting, while the MSE was calculated from 10 different sets of noisy data. While all models fit the general shape of the true data, we are surprised to note that the Log-Probit model did not best recover the original data. Even though a Log-Probit model was the source of our data, our results show that the addition of noise may result in the best-fitting model being one other than the original functional form. However, even the worst-performing parametric models in terms of MSE (Probit and Weibull) were not much worse than the best-fitting models, differing by less than one order of magnitude. This is evidence of the ability of this class of models to fit the general shape of a risk function when the original curve is derived from one of the member functions.

Table 4: Simulation: Mean Squared Error

Model	MSE
Logistic	0.002
Probit	0.010
Quantal-Linear	0.005
Quantal-Quadratic	0.007
Multistage (Two-stage)	0.004
Log-Logistic	0.008
Log-Probit	0.007
Weibull	0.008
Dichotomous Hill	0.001
Bayesian Model Average	0.259
Gaussian Process	0.002

4.1.2 Bayesian Model Averaged Results

To fit a curve that accounts for model uncertainty, we produced a BMA curve by weighting each estimate from our parametric model section by the posterior probability of each model. Even though the Dichotomous Hill model was the best-performing model in terms of BIC and MSE, it has a relatively low posterior probability. The models that contributed most to our BMA function were the Logistic and Probit models. Again, this is surprising as the data was generated from a Log-Probit model. However, after multiple runs, these models retained the highest marginal likelihood. Additionally, even though some models contribute less, all models contribute to the final model.

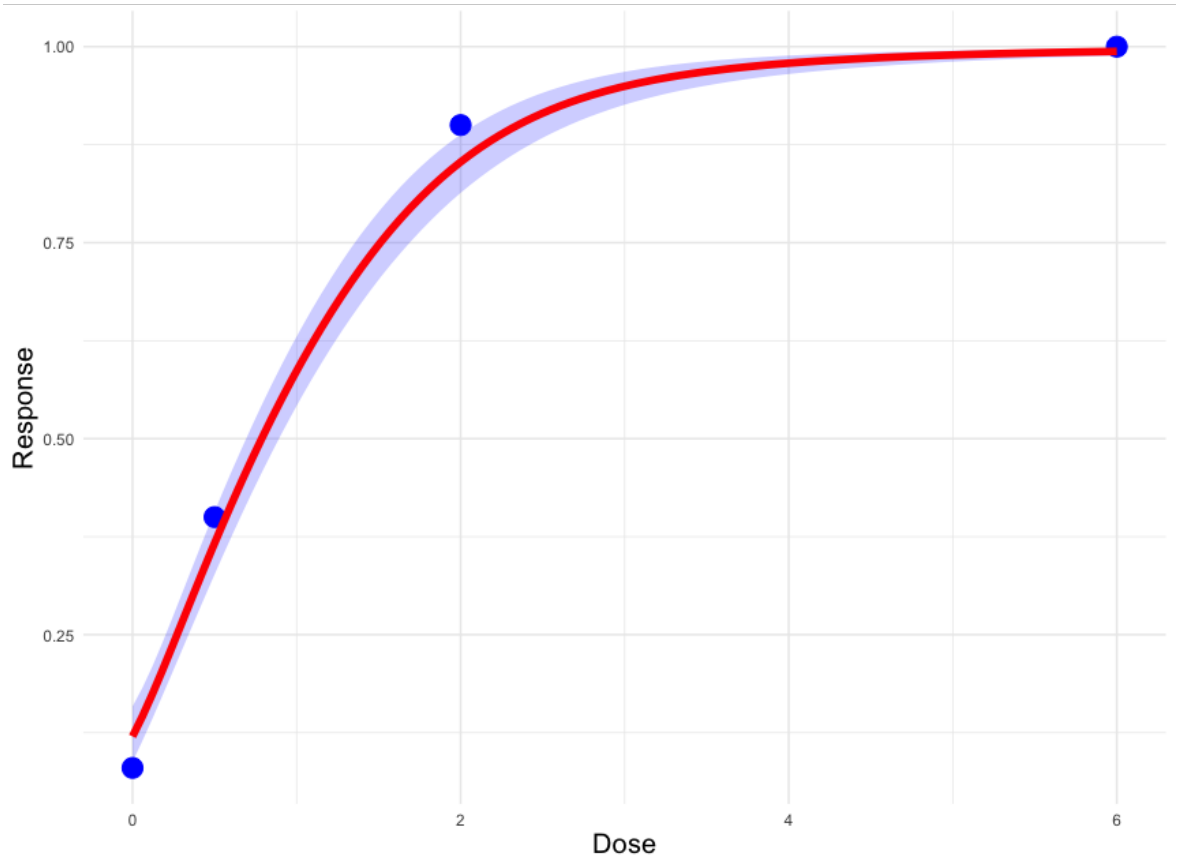


Figure 2: Simulation: Bayesian Model Averaged Dose-Response Curve

Figure 2 shows the BMA results plotted alongside the true observations. By visual inspection, the curve appears to resemble the shape of the best-fitting model, though the credible intervals are narrow across the function. These results differ from examples in the literature, where BMA results have differed from the single best model. However, it is unsurprising that the parametric models are able to recover an underlying curve from the same class of biologically-plausible models. Despite the desirable shape of this curve, the MSE value for the BMA function was the worst among all tested models, and the credible intervals did not capture two of the observations. The tighter credible intervals may result from increased confidence at higher levels, particularly as the highest dose was associated with consistently high response levels for all component models.

Regarding the MSE performance, this may be an artifact of our coding method to save compu-

tational resources. We calculated the marginal likelihoods of the models based on a single run of each parametric model. We then used these marginal likelihoods to derive our posterior probabilities. Even though we used a mean of the parameters from the 10 iterations, our weighting of the component models remained the same for the 10 iterations of fitting noisy data. In this way, the BMA is still testing 10 sets of observations with added random noise, but it is not running the complete set of 10 fittings of the nine parametric models and 10 separate marginal likelihoods (and subsequent posterior probability) calculations. This method reduces the run time of the BMA estimation but also means that the BMA model is being tested against a more challenging standard of trying to approximate the curve based on one set of realizations of model weights.

4.1.3 Gaussian Process Regression Results

The GPR was among the best-performing models in terms of MSE, and its curve showed good fit with the data, such that the estimations are practically coincident with the observed values. (We double-checked to ensure we fitted the model to the noisy data and plotted the original data.) Again, we are showing the curve's results from only a single realization of noisy values in Figure 3, plotted alongside the true observations. GPR demonstrates a similar shape to the best parametric models and the BMA results while capturing all of the data points in its credible intervals.

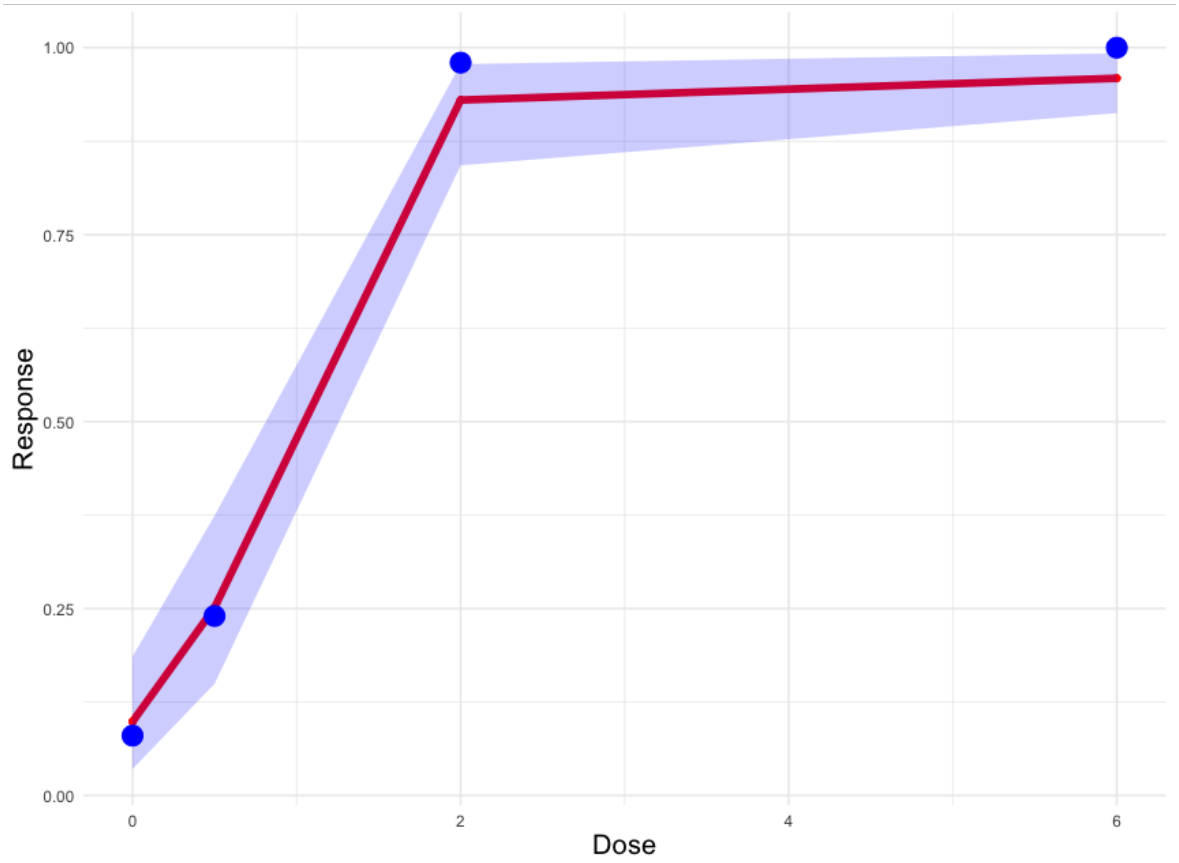


Figure 3: Simulation: Gaussian Process Dose-Response Curve

4.1.4 Diagnostics

Regarding diagnostics, all simulations showed good convergence when we examined the ACF, trace plots, and pairs plots. Parameters in all models demonstrated Rhat values of less than 1.02, with most Rhat values of 1.00.

4.2 Antimony Results

4.2.1 Bayesian Parametric Results

Using the methodology previously outlined, we fit the antinomy data to the nine parametric models.

Among the parametric models, the dose-response curves produced by fitting the Log-Probit, Log-Logistic, and Dichotomous Hill models offer the only visually-reasonable approximations of the data from the model space we explored. In Figure 4, we present the nine models with credible intervals for each estimate, alongside the observed data. Further supporting these models, the Log-Logistic and Log-Probit curves both capture all the observed data points in their credible intervals, and the Dichotomous Hill model captures three out of the four data points in its credible intervals. Based on the standard methods of comparing models in this form of parametric assessment, we found that the Log-Probit model demonstrated the lowest BIC (Table 5). In the parametric approach, this model would typically be the foundation for our BMD and risk function estimates. While the Log-Probit model had the lowest BIC, the Log-Logisitic model offered a nearly-equivalent BIC value. Without clear separation in these two values, the model choice of the Log-Probit model over the Log-Logistic may result in the loss of data about the error presented by another similarly-performing model.

Outside of the three models discussed above, the other models do not adequately capture the data's shape. These results are markedly different from our simulation study, where all models produced a plausible fit.

4.2.2 Bayesian Model Averaged Results

The BMA dose-response curve (Figure 5) appears to approximate the best individual dose-response curves, particularly the Log-Logistic, Log-Probit, and Dichotomous Hill curves. However, we can see that the model with the highest posterior probability is the Log-Logistic model, differing from the best-performing model according to BIC (Table 5). While the difference in BIC between the two models is small, the difference in their posterior probabilities is large. However, both models fit the data reasonably well and are incorporated in our BMA results. Together, the Log-Logistic and Log-Probit models contribute over 99% to the weights when averaging over the models. This again differs from the simulation, where most models contributed at least 9%, with only two models contributing 2% or less. The antimony data suggests a concentration of the weights between the two models. This reduces the variation in the functional forms. One of the aims of BMA is to incorporate various error structures to support a better understanding of uncertainty in model choice, but this benefit may be limited given how mis-specified most of these models are for the given data. However, the BMA results also demonstrate greater confidence in its estimates in the regions where the models overlap. The model consensus we see across our best parametric forms reduces uncertainty in the overlapping regions while largely discounting the uncertainty of estimates in poorly specified models (such as the Quantal-Quadratic model).

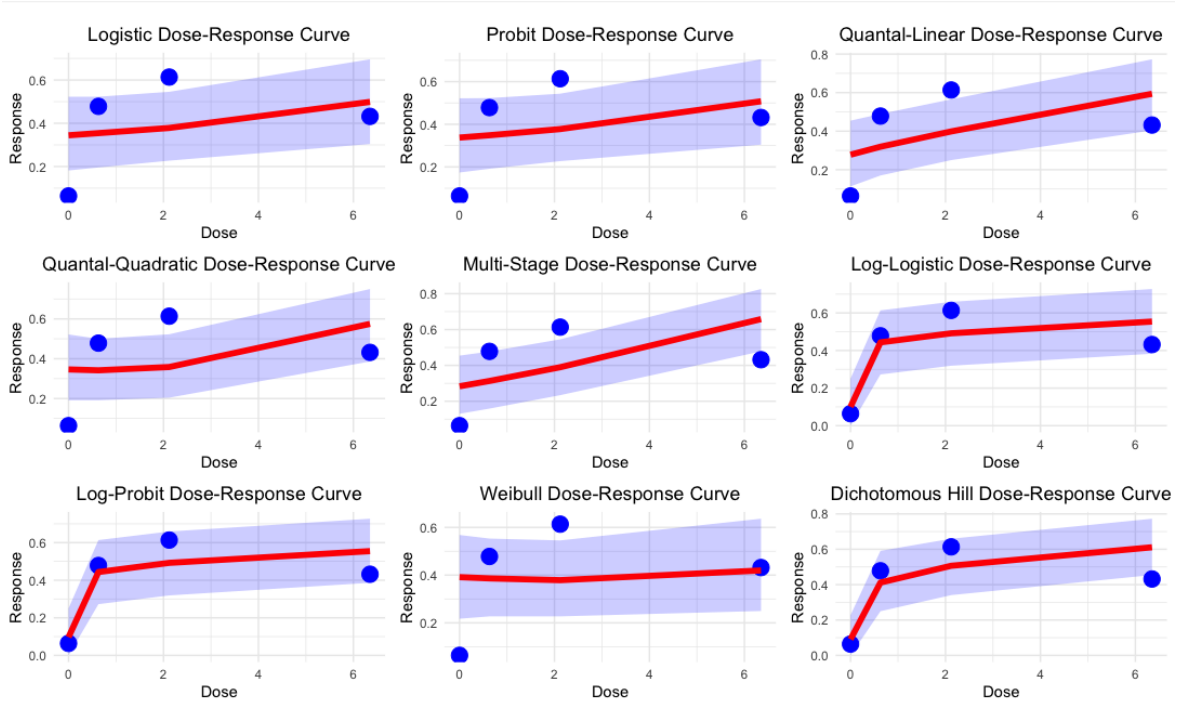


Figure 4: Antimony: Bayesian Parametric Dose-Response Curves

Table 5: Antimony: Parametric Model Comparison

Model	Marginal Likelihood	Posterior Probability	BIC
Logistic	-121.285	0.000	53.452
Probit	-122.784	0.000	53.507
Quantal-Linear	-124.124	0.000	59.291
Quantal-Quadratic	-135.252	0.000	52.017
Multistage (Two-stage)	-140.703	0.000	60.177
Log-Logistic	-110.334	0.762	28.198
Log-Probit	-111.504	0.236	28.174
Weibull	-122.033	0.000	57.651
Dichotomous Hill	-116.190	0.002	31.533

With this greater confidence in the estimates, the credible intervals of the BMA results are narrow and fail to capture two of the four data points. (The lowest dose is captured by the credible intervals, but a problem with the Weibull model at that dose has caused an issue with visualization. We will comment further on the problems with the Weibull model). The mean prediction values appear reasonable, but the tighter error bounds do not capture the data.

4.2.3 Gaussian Process Regression Results

The GPR also approximates the best-fitting parametric models but without the parametric form constraints, which bolsters our confidence in the general shape of the results.

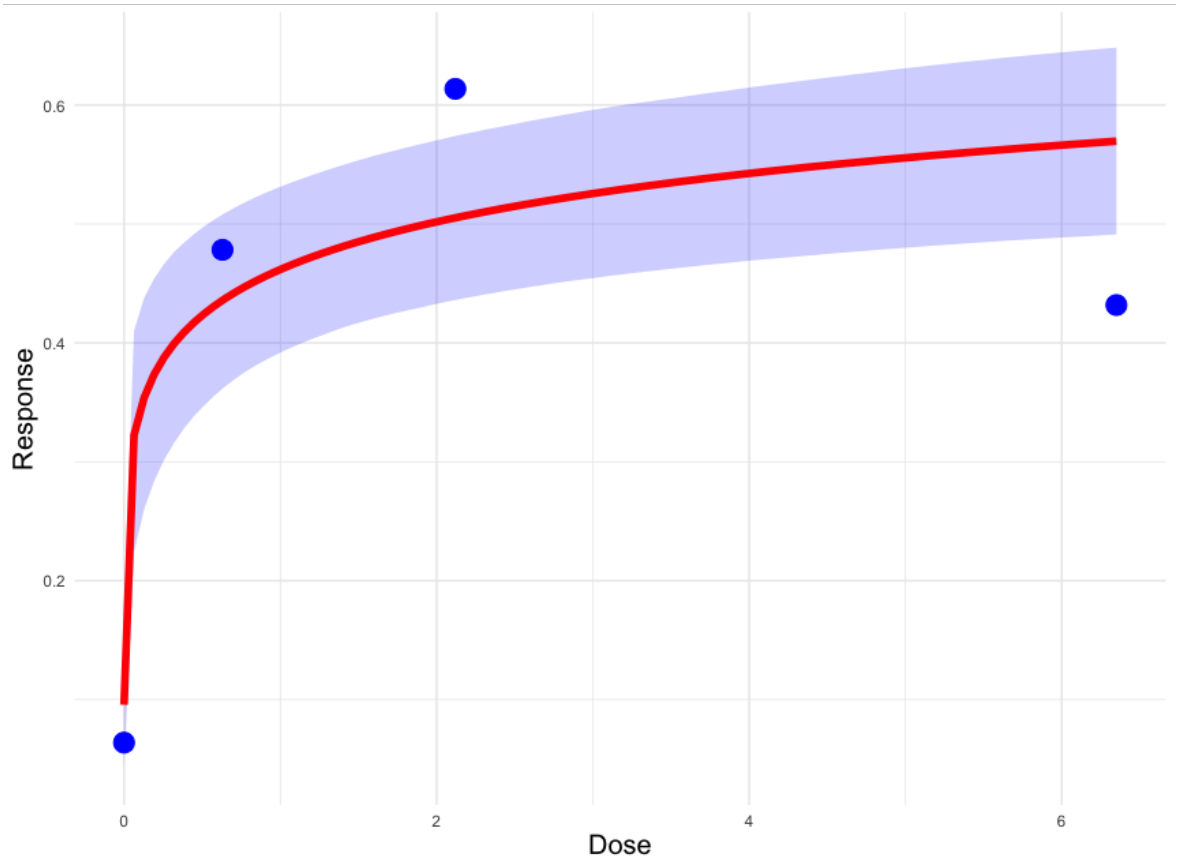


Figure 5: Antimony: Bayesian Model Averaged Dose-Response Curve

As shown in Figure 6, our GPR results capture three of the four data points in the credible interval. With only monotonicity and bound constraints, the GP has approximated the shape of the function in a manner consistent with expectations and in line with the outcomes of the other methods. However, the credible intervals of the GPR fail to capture the final point.

4.2.4 Diagnostics

With one exception, the diagnostics were adequate with respect to ACF, trace plots, and pair plots. The Rhat values were 1.00 for all parameters. The models all demonstrated good convergence and low autocorrelation after the burn-in values. One model in our Bayesian parametric section did present diagnostic challenges, the Weibull model. While the Rhat values were all 1.00 and the effective sample size was high, the other diagnostics indicated an issue. The ACF plot did not drop sufficiently across the three parameters in all chains, showing higher than desirable levels of correlation among samples. The trace plots indicate issues with our β_0 and β_1 parameters, potentially poor mixing. Most concerning were the pairs plots, which showed dependence between the parameters. Increasing the chain length and exploring different initial values did not resolve the problem. Initializing was also an issue for this model. As such, the problem may be with the priors, specifically for the β_1 parameter, as this is a semi-informative prior from the EPA's BMDS that is attempting to use the lognormal distribution to encourage prior samples of greater than one. However, this distribution

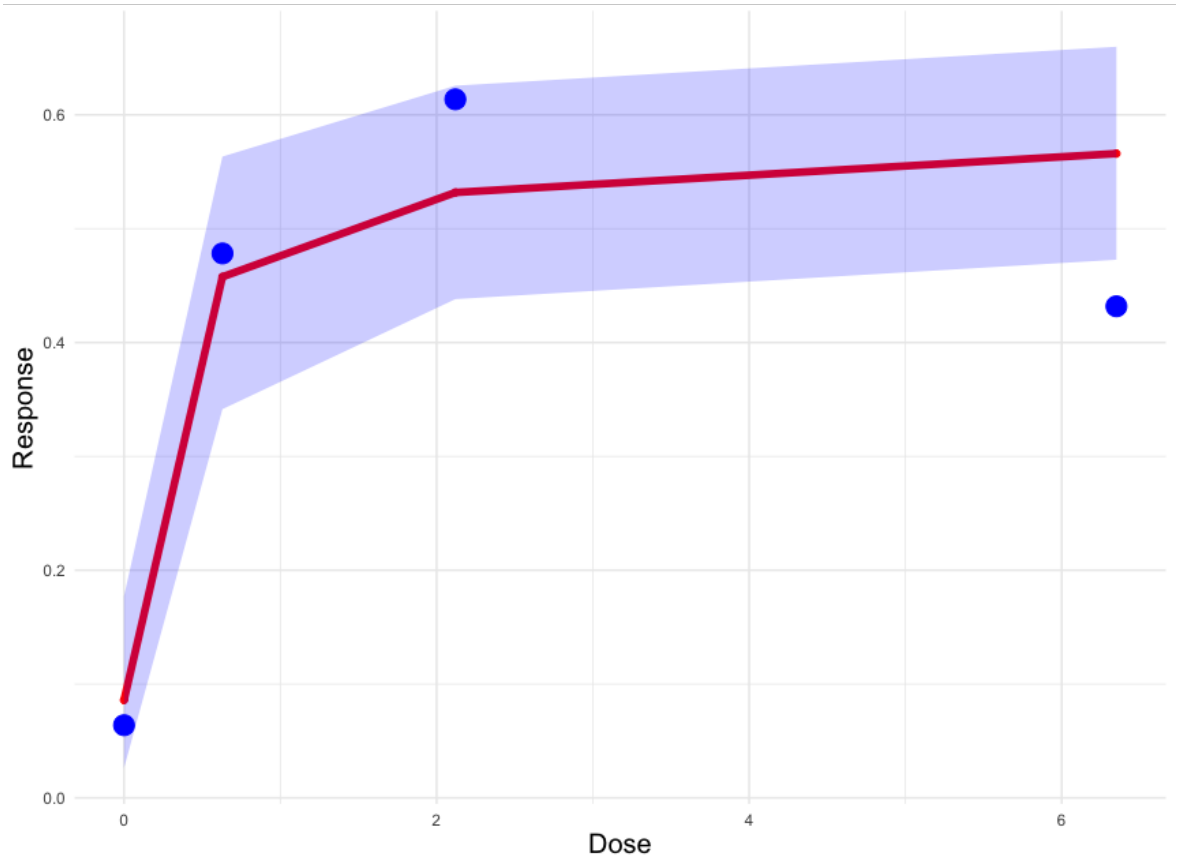


Figure 6: Antimony: Gaussian Process Dose-Response Curve

may be suboptimal or inaccurate, and other distributions should be explored. We may also have an issue with this specific parameterization of the model and how it addresses this particular data set, as we did not have this issue with our simulation data using the same prior. However, we also did not encounter this problem while fitting this model while using a Normal prior on that parameter. It is likely a combination of the prior, data, and parameterization causing the issue. In future work, we will explore alternative parameterizations and priors to better accommodate non-monotonic data.

5 Discussion

5.1 Priors

5.1.1 Effect of Changing Priors

Given the limited number of observations in toxicology studies and the limited number of original experiments for many agents, prior belief may strongly influence the posterior for the parameters and, as a result, the dose-response curve.

We explored alternative priors from the literature, such as inverse gamma priors on the [0,1] constrained parameters, Normal priors with tighter bounds (i.e., Normal (0,1)) on positive values (also only accepting positive draws) (Fang, Piegorsch, and Barnes, 2015; Fang et al., 2015), and uniform priors on all variables across either the [0,1] range for bound parameters or a wider range such as [-40, 40] for unconstrained parameters as in Shao and Shapiro (2018). The estimates and credible intervals differed among the parametric models, but the shape and general model ordering remained consistent.

Despite this, we do not have an adequate understanding of the effect of changing priors on limited experimental toxicology data from the values we explored. More rigorous sensitivity analyses are warranted to understand the magnitude of the effect of the priors on our different methods.

5.1.2 Prior Elicitation

This research may be improved with better prior knowledge of both the parameters of our parametric models and an understanding of prior probabilities for our models under consideration. As such, further research should explore informative priors for specific classes of agents, both for the parameters and models.

While our current values are based on recommendations from the U.S. EPA's recently-released prior recommendations (2022), those recommendations appear constrained by a coding environment because they impose nonnegativity constraints using the prior rather than specifying that constraint in the code to reject negative samples. This suggests these priors may not be the best tuned for purpose.

These priors could be improved further based on classes of agents rather than general guidance. By examining past results for dose-response curves among a sufficiently-large sample of a given class of toxins, we may be able to identify priors on our parameters and models in the parametric and BMA sections. For BMA, we anticipate that specific models generally fit a given set of agents (such as pesticides with a specific mechanism of action) better than other models. Another method for setting priors for the models may be to return to an older paper. Good (1950) suggested starting with a uniform prior and updating the prior using "imaginary data" generated by domain experts. The posterior of this experiment would be the prior distribution for the model space. While more involved, this offers another method of incorporating domain expertise into modeling where limited experimental data are available.

5.2 Model Inclusion

Model averaging helps researchers address the problem of choosing the best model among a set and reduces uncertainty in estimating the dose-response relationship. However, this does not eliminate

the challenge of model selection, as we must still consider a set of models for inclusion. We can still question whether the class or group of models under consideration reflects the actual biological process and whether we have models that degenerate into other models that are then double-counted.

As better models are identified, those that are no longer biologically-consistent or which fail to explain the data may be replaced, or new models can be added without replacing the old models. In 1994, Bernardo and Smith discussed the difference between two perspectives on model selection: the case in which all the classes of models under consideration are known in advance, which they called the "M-closed perspective," and the case in which all the models are not fully known in advance, but may require additional iterations or analysis to elicit, which they called the "M-open perspective."

Hoeting et al. (1999) built on this discussion, stating that while BMA initially appears to be a form of "M-closed," in the more abstract sense, BMA can incorporate additional models as new, relevant models are identified (Hoeting et al., 1999). Exploring a larger space of models may better account for the uncertainty inherent in model choice. However, the new models should accommodate new forms of curves not yet covered by the existing models. The method of BMA we present can accommodate a more extensive range of reasonable models as they are identified.

5.2.1 Model Space Reduction

BMA remains computationally demanding. While we attempted to reduce these demands with bridge sampling, the estimation and computational demand for even a limited set of data is still non-negligible. To reduce the computational demand of the BMA method and prevent aberrant behavior when combining the models, we can set thresholds on posterior probabilities so that we only average over models with greater than some threshold of posterior probability. This would require additional theoretical examination, and would serve as an extension of suggestions from Raftery et al. (1996). (For example, we may consider including only the models that have more than a 1% posterior probability and that "discarded" probability may be distributed proportionally (based on their relative probabilities) between the remaining models).

Additionally, we may use a simple selection mechanism (such as BIC) to identify a set of best-performing models and then use BMA on this more limited selection. Further research should explore methods capable of evaluating a larger set of unique models while only averaging over those that make sufficient contributions to be worth the computational resources necessary for the marginal likelihood and posterior probability estimation and output combination over the risk function range. Research should also explore the effect of such methods on the risk function.

5.3 Model Performance in Varying Contexts

5.3.1 Simulation vs Antimony

Our findings show a noticeable difference in model performance between the simulation and antimony datasets. While most of the parametric models in the simulation component offered a visually-adequate fit and made at least moderate contributions to the BMA curve, in the antimony component, only three of our models fit, and only two made moderate contributions to our BMA results. As such, these models may not have been sufficient to explore the range of functional forms for dose-response curves.

The ceiling in outcomes from the simulation set may partially explain the difference between the simulation and antimony datasets. As the simulation set had almost all subjects experiencing the outcome in the highest dose groups, this likely reduced the challenge of fitting those final points and made it more likely that the true data would fall within the credible intervals. Conversely, in the antimony dataset, fewer than two-thirds of the subjects in the higher dose groups demonstrated the outcome, making dose-response estimation more challenging, and this was further compounded by the non-monotonic nature of this particular set.

5.3.2 Parameterization

Previous research has suggested reparametrizing models around BMD estimation rather than the individual parameters. From this, researchers directly estimate this value of interest rather than the dose-response curve. While we are mainly interested in the broader dose-response curve, this type of parameterization remains compelling for how relatively intuitive BMD is and for the potential sharing of BMD values across toxins of a similar class. We look forward to exploring a reparameterization in terms of BMD in future papers (rather than solving for BMD) to apply prior information that may be more commonly shared across agents of the same class.

Furthermore, we plan to consider hierarchical models in our dose-response assessments, such as those that separate the effect of male and female rodents. We may extend this to incorporate human data into the estimates to share power across multiple studies while exploring varying dose levels.

5.3.3 Initialization of Parametric Models

In the results presented, we did not set initial values but let R initiate its starting values for the MCMC chains. We made this change after updating our priors to the presented settings. With our previous Normal (0,10) priors on all parameters not constrained to [0, 1], we set reasonable initial values, and the MCMC sampling ran as expected. However, using lognormal priors with the same initial values, the chains in some of the models could not initialize and, as such, were not able to sample from the distribution. Therefore, we removed our initial values, and R could once again sample the space. Allowing R to set the initial values may result in poor reproducibility. We will seek to understand how and why the lognormal priors influence the initialization of the MCMC chains in future research.

5.3.4 Full vs Partial Data

In the parametric, frequentist BMD analysis of the NTP antimony dataset by Schildroth et al. (2022), the researchers removed the highest dose group when modeling with the subset of parametric models for dose-response estimation, a method accepted by the EPA but one which comes at the cost of data. Removing the final data point because response levels are lower than expected (often due to high mortality from alternative causes) may be applied to improve the fit of the parametric models at the lower doses, which is typically the area of interest in BMD estimation. Conversely, all our methods include all data points.

Future research could explore the performance of our methods when evaluating a range of datasets, modeling the complete data and also the data minus one dose level (such as removing a single dose level at a time until all possible models have been fitted) to understand the effect of data removal from

each technique. However, as we are interested in how the model performs across the range of tested doses, understanding the modeling implications of data removal at higher doses may be helpful for dose-response estimation across the broader risk function and understanding which method results in the most accurate and stable estimations (when true or well-validated risk curves are available).

The response at the final dose level posed significant challenges to all models. This final point may lie outside the normal range of expectations based on other factors related to the experiment. Even though the development of specific cancers may be stochastic, in this particular experiment, large numbers of mice died in the highest dose group. Premature death in the highest-dose group removed a large number of mice from the risk of developing cancer and, therefore, the outcome count. Despite this challenge, it would still be preferable if our models could accommodate such variation in the data.

5.4 Noise and Experimental Design

The antimony dataset we considered (and cancer counts in general) is unlikely to follow a discrete uniform noise function, particularly one in the [-3, 3] range. If the noise is generated by a uniform distribution, then the range is likely much larger than what we applied given the antimony observations. Future research should explore better methods for modeling noise in cancer outcomes (or other biological processes of interest).

We should also explore the noise limits at which our methods fail to recover a reasonable dose-response curve. We plan to explore whether the parametric, BMA, or GPR approaches offer a superior method to recover that underlying function at higher levels of noise, to understand at what level of noise is the recovery of the underlying function no longer possible (including range and also distribution of noise), and how the function behaves before degenerating. This may help researchers better model the underlying process.

5.5 Gaussian Process Regression Considerations

5.5.1 Discrete and Continuous

The GP model appeared to approximate the underlying functions well, with fewer computational resources than the BMA approach. GPR was a valuable method for checking our parametric construction. Additional research that explores a continuous version of this discrete GPR is warranted to obtain a smoothed function that better displays inflection points across the range of exposures and allows for better interpolation between observations.

5.5.2 Kernel Selection and Tuning

We employed the RBF kernel for our GPR due to its application in past literature and good visual fit. However, there are many other kernels we could have applied, alongside combinations of kernels. We also experimented with some pre-built GPR functions (without optimized hyperparameters) with various kernels, including the Polynomial, Rational Quadratic, Dot Product, and White Noise Kernels. These all performed poorly in fitting the data. Future research should explore the development or use of other kernels to understand which kernel demonstrates the best properties for fitting this form of data.

Even within the RBF kernel, it is possible to tune the hyperparameters in ways that change the results of the dose-response curve. Even though there are only two hyperparameters in this kernel, these could be set to a wide range of values that could be reasonably supported. While we used parameters that maximized the marginal likelihood of the model, it may also have been possible to use other, larger datasets to set hyperparameters and then use those values for estimating the risk function from our antimony dataset. While this may be appropriate if there is a reasonable foundation for shared hyperparameters among different toxins or classes of toxins, this may also be used to change the shape or credible intervals of the function.

Before building our GPR, we also built an approximation of a GP with a random walk prior method. When we changed the priors on our initial value from a Normal to a lognormal distribution, the shape of our function changed, and our risk at lower levels increased from the current estimates to levels nearly at the risk levels of our 0.63 mg/kg-bw/day levels. This degree of change in baseline risk could meaningfully alter excess risk calculations and have serious health, policy, and industrial impacts. Further sensitivity analyses are necessary to understand the effect of changing hyperparameters and kernels when conducting GPR with limited data.

5.6 Additional Considerations

5.6.1 Computation

Regarding the computational demands of the models we considered, the BMA approach required the most resources. This approach required fitting every parametric model, computing the weights for each model, and then summing based on the weights. The GPR took the least time to run, even with the demands of inverting multiple matrices. Given the relatively limited experimental data in standard toxicology studies, GPR requires lower computational resources than fitting the nine parametric models. However, as the size of our inputs grows, so too will GPR increase in computational demand. Further research should explore the point at which GPR would be more demanding, which may occur as we expand to epidemiological studies.

5.6.2 Monotonicity

While we assume the risk function is monotonic, this is not always true. In some instances, low-dose exposure to an agent may be associated with a decreased risk of an outcome, with risk then increasing at higher doses. Future research should explore whether the modeling of the risk function should consider the possibility of low-dose protective effects. As this research seeks to explore the effects of agents (typically expected toxins) on the development of an adverse health outcome, it may be reasonable to maintain the monotone constraint when conducting a toxicology assessment.

However, in the case of testing therapeutic or potentially-therapeutic agents, additional research is warranted to explore the potential of separate model fitting for lower doses or a combined method to account for protective effects at low doses. (This area may be addressed by exploring the current methods for evaluating public health interventions, such as folic acid in bread or chlorine in water.)

5.6.3 Risk Function Interpretation and Manipulation

Considering the interpretability of the models, communication of BMD and BMDL estimates to relevant stakeholders would likely remain unchanged. However, the process of arguing and setting these estimates may become more difficult given the greater complexity of the Bayesian methods compared to frequentist methods. The more technical risk assessment process may become increasingly complex when considering Bayesian parametric models, BMA, and GPR as the sources of estimates. This makes interpretation – particularly as we explore varying methods to understand the risk function across an entire range of relevant exposures rather than a single PoD – more difficult.

These methods also open new possibilities for risk function manipulation. Priors on the parametric model parameters, assumptions in setting priors over the models, and kernel choice are all starting points for motivated groups to challenge results or inject their preferences into recommendations. The selection of a seed in MCMC sampling, the method for estimation of the marginal likelihood, and the tuning of GP hyperparameters could become tools to produce tailored outcomes.

Additional research that evaluates these methods against known functions from real-world data may allow for greater confidence in the methods. If these methods estimate true risk better than existing methods, they may be accepted in policy and industry despite the additional complexity. However, if the models are overly conservative, insufficiently protective, or malleable to tuning in a way that fails to reflect the underlying model, they should not be applied. As such, the stability of these models to intentional changes is an area that merits further research to guard against manipulation.

5.7 Broader Questions

This initial exploration of Bayesian risk function estimation also highlights a series of broader questions about developing a comprehensive and systematic approach to Bayesian human health risk assessment.

Beyond the scope of this initial paper on Bayesian risk function estimation, we have identified a series of additional questions as we seek to create a systemic approach for Bayesian human health risk assessment. We intend to explore how dose group sizes influence confidence in estimates and GP optimization to conduct more efficient animal toxicology experiments, the allocation of subjects into potentially more dose groups, the sharing of data across animal and human data studies in a hierarchical fashion that naturally elicits a conversion factor, the use of these methods for low-dose extrapolation, and the application of these methods (with potentially different parametric forms) to the analysis of drug development.

Our analysis of the antimony data focused on the most sensitive species and sex in the experiment, consistent with standard practice. However, a method that explores both sex groups may be helpful in estimating the true risk function for male and female mice, which appear to be differentially impacted by antimony – the same may be true for other toxins. Additionally, if there are multiple reasonable animal models, the inclusion of these different species may reduce the impact of specific biological characteristics found in one animal that may bias the estimates. At the same time, this may increase our information about the agent. A method for incorporating additional observations, potentially through models that allow for greater information sharing across nested groups may also be useful.

Models that can combine multiple species, both sex groups, and both epidemiological and exper-

imental toxicology studies may support uncertainty reduction in species conversion and uncertainty factors related to data quality. This major undertaking will require considerable additional research to show superiority to current methods.

6 Conclusion

In this study, we explored three different Bayesian methods to estimate the risk function in the dose-response assessment component of the human health risk assessment process: Bayesian parametric model fitting for a set of biologically-plausible models, BMA over the subset of plausible models, and GPR with monotonicity and bound constraints. We applied these methods to a simulated dataset and an experimental antimony dataset.

6.1 Simulation Conclusion

The results from our simulation component demonstrate that the three Bayesian approaches we present can reasonably recover the dose-response relationship when moderate noise is added to data derived from a known parametric function.

Within our Bayesian parametric approach, all models offered a reasonable fit, but the Dichotomous Hill model offered the best-fitting model based on BIC, with also the lowest MSE across multiple simulations.

The BMA approach produced a shape similar to the best-fitting parametric model, though the error bars were narrower than the parametric approach, indicating lower levels of uncertainty around the estimation. Furthermore, all models contributed to the model-averaged curve, with most contributing 9% or more. This method reduced the risk of overfitting a model, as the parametric approach recovered a reasonable shape but the wrong model.

The GPR method offered superior performance to the other two methods, offering an excellent visual fit and a very low MSE. The GPR approach's flexibility and capacity to capture the data points within its credible intervals demonstrates the method's ability to approximate complex dose-response curves with limited constraints.

6.2 Antimony Conclusion

Table 6: Antimony: Three Methods Estimate Comparison

	Dose levels (mg/kg-bw/day)			
Method	0.00	0.63	2.12	6.35
Parametric	0.097 (0.00-0.24)	0.443 (0.27-0.61)	0.493 (0.33-0.66)	0.556 (0.38-0.73)
BMA	0.096 (0.04-0.18)	0.437 (0.36-0.51)	0.505 (0.44-0.57)	0.569 (0.49-0.65)
GP	0.086 (0.03-0.18)	0.458 (0.34-0.56)	0.532 (0.44-0.63)	0.566 (0.47-0.66)

Across all three methods, the mean risk function estimates were consistent across the tested dose levels. As shown by Table 6, the estimates were relatively similar, with no one method offering clearly conservative or non-conservative mean estimates relative to the other methods. However, it is notable that the Bayesian parametric method (represented by the Log-Probit model) had the widest credible intervals at each point, indicating a higher level of uncertainty when using this method. The use of the Bayesian parametric approach and BMDL as a PoD (rather than BMD) would result in a more conservative assessment of risk. Therefore, while the models have similar estimates, the

parametric method in this example produces more conservative estimates when using BMDL as the PoD. This method was the only one in the antimony analysis to capture all the data points within the credible intervals. Despite these attractive results, only three of the models in the parametric approach offered a reasonable visual fit to the shape of the data.

The BMA method reduced uncertainty and produced a plausible dose-response curve but failed to include two observed data points within the credible intervals. The BMA risk function estimate was again similar in shape to the best-fitting parametric model, but only three of the nine models had contributions of greater than 0.1%, (and with the vast majority of contributions from only two models).

The GPR method balanced flexibility and fit, capturing most data points in its credible intervals but still missing one at higher doses. This additional method bolsters our confidence in the shape of the curve.

These methods and data indicate that exposure to antimony is associated with a steep increase in the risk of developing cancer among mice between the 0.00 and 0.63 mg/kg-bw/day levels, and then the increase in the rate of cancer risk reduces around this value to a more gradual increase. This suggests that higher doses of antimony trioxide exposure (above 0.63 and up to 6.35 mg/kg-bw/day) may not be an efficient area for influencing cancer risk. Rather, health work should focus on preventing exposure at lower doses. If higher doses are necessary or unavoidable, resources may be better spent on other toxin exposures rather than attempting to reduce exposure from 6 to 2 mg/kg-bw/day, for example.

These results also indicate that the cancer risk may not be linear across the range of exposure. At lower doses, it may be reasonable to assume a linear, no-threshold model, but at higher doses, these results may not hold. (However, premature death in the higher-dose groups may also bias these results.)

6.3 General Conclusions

Among these methods, the Bayesian parametric model fitting method showed the most uncertainty in its estimation and managed to capture the most observations in our data sets. This indicates that while single parametric models are simpler and more prone to larger credible intervals, they may better capture the variation and uncertainty in estimates across the dose-response range.

In contrast, the BMA approach, by averaging models with the highest posterior probabilities, presented a curve with reduced uncertainty around the estimates and a plausible form. However, this reduction in uncertainty sometimes led to credible intervals that failed to capture observed data points. BMA was also the most computationally intensive method among the approaches we explored.

The GPR method most flexibly fit the data, with only limited constraints. This method effectively captured most of the data points in its credible intervals. Despite its benefits, the GPR method makes it challenging to incorporate easily-interpretable prior information, especially in terms of biologically-meaningful parameters like BMD. Further research is required to understand if similar toxins share behaviors associated with specific kernels and hyperparameters. While this model was relatively easy to implement and required limited computational resources at this scale, gaining support for its use in a policy context may be challenging given the ability to manipulate hyperparameters, kernels, and seeds to influence results.

Our results differed from previous papers, with the parametric models offering inconsistent fit to the real-world data and being the most conservative models in our study (if assessors apply BMDL as the PoD). Our BMA results were similar to our best-performing models, without major differences in shape at lower doses. Finally, our GPR with monotonicity constraints represented the dose-response curve that was consistent with our other models (without being more conservative than other methods).

Our preferred method for producing a risk function is the BMA method. Though our worst-performing method in terms of MSE (for the simulation) and in capturing the data points (with only two of the four antimony data points in the credible intervals for the antimony dataset), the BMA method offers the most theoretically desirable qualities, more easily incorporates prior information on the individual model parameters, constrains the model with plausible functions to avoid overfitting, and allows for prior probabilities on the model space. The BMA approach is more comprehensible than GPR for application to health, policy, and industrial operations. It reduces concerns regarding single model selection (in the parametric case) and kernel selection and tuning (for the GP case). Finally, the BMA approach is open to the inclusion of additional, plausible model forms as our understanding of biological processes and systems evolves. As such, the BMA approach can incorporate additional methods that will help combine prior knowledge with data in a balanced way to estimate the risk function of abiotic agents.

We recommend the use of BMA that considers a set of biologically-plausible models as the method for discerning the risk function in the dose-response component of human health risk assessment. We also suggest the inclusion of GPR to confirm the shape of the BMA curve. Future research should refine the selection of models under consideration, enhance the incorporation of priors, establish methods for comparison between the BMA and GPR results, and explore additional experimental and computational methods to increase the speed and accuracy of risk function estimation.

7 References

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